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(54) **SUCTION MOUNT AND BRACKETS AND ACCESSORIES THEREFOR**

(71) Applicant: **Olson IP Technologies, Inc.**, Mundelein, IL (US)

(72) Inventors: **Erik Olson**, Deerfield Beach, FL (US); **Richard Carl Olson**, Deerfield Beach, FL (US)

(73) Assignee: **Olson IP Technologies, Inc.**, Mundelein, IL (US)

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See application file for complete search history.

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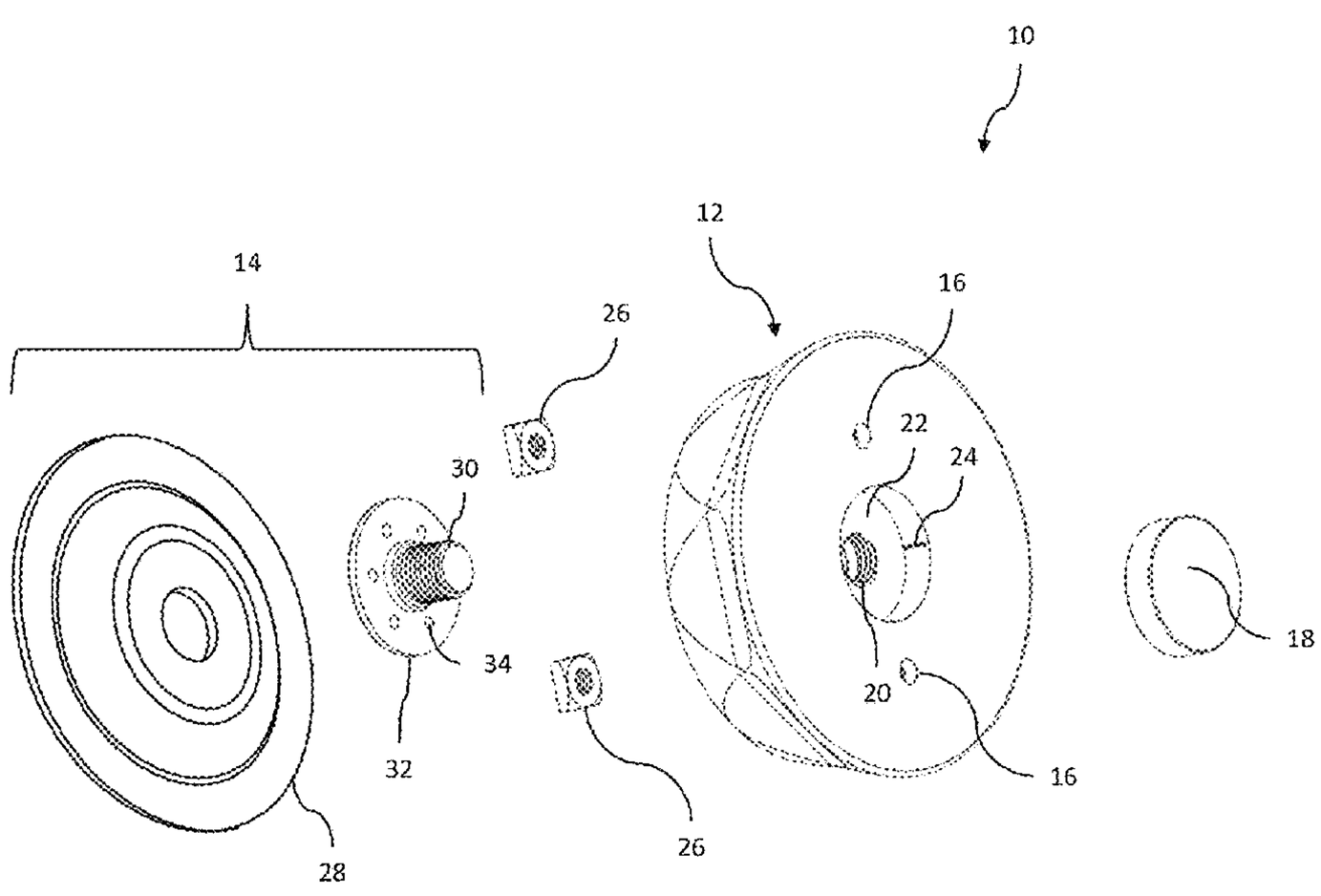
Primary Examiner — Monica E Millner

(74) *Attorney, Agent, or Firm* — Fitch, Even, Tabin & Flannery, LLP

(57) **ABSTRACT**

Unibody suction mount having a compact design allows for a plurality of products to be fastened onto the mount. The mount portion includes an integral threaded portion that cooperates with suction cup having a threaded post. The mount includes direct engagement with the suction cup to hold the suction cup in place as the vacuum is being created in the suction cup. The suction mount utilizes one or more nuts to allow for attachments to be fastened onto the suction mount. The attachments may include a variety of brackets used to connect a variety of accessories. The accessories can be designed for specific activities.

35 Claims, 25 Drawing Sheets



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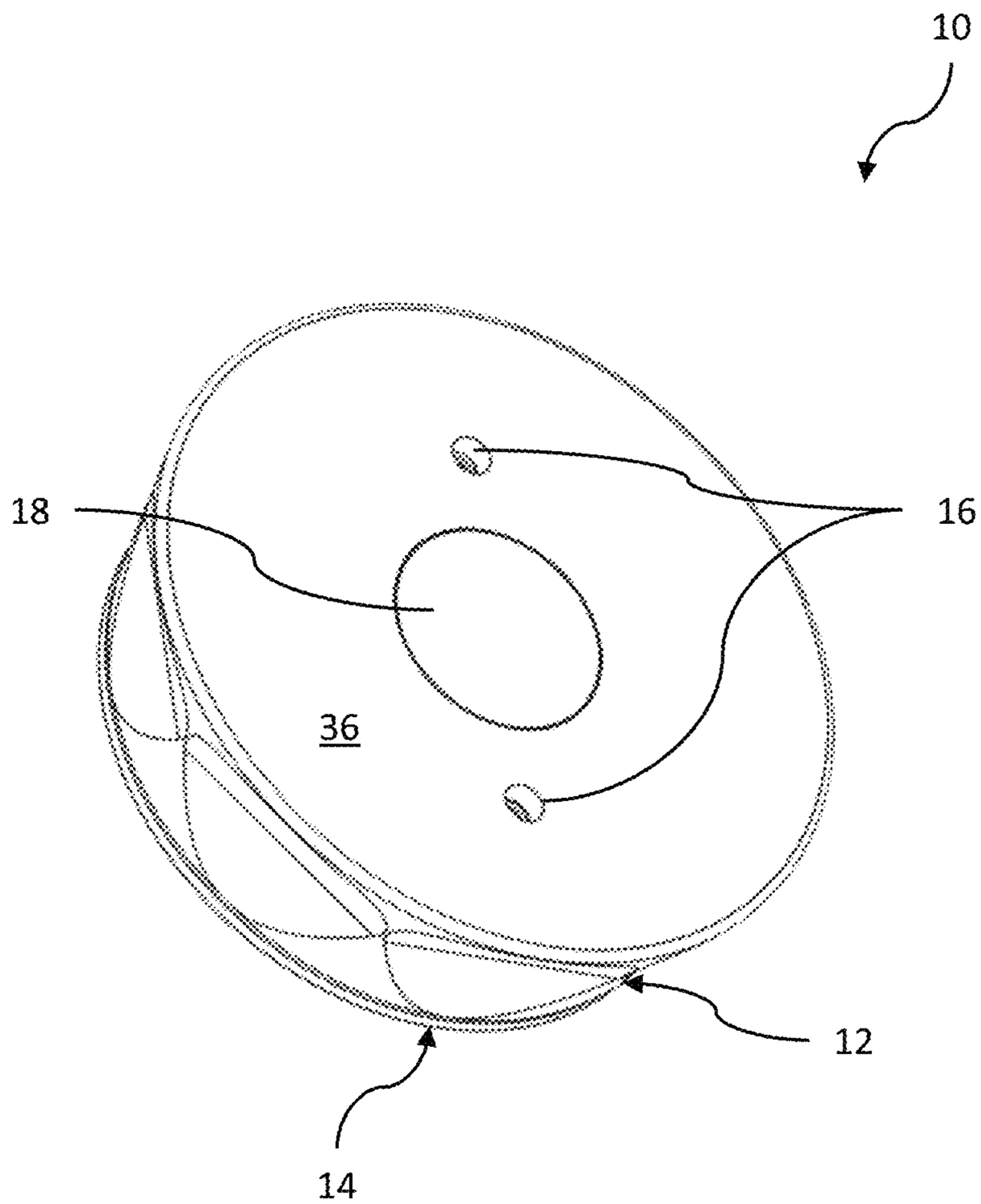


FIG. 1

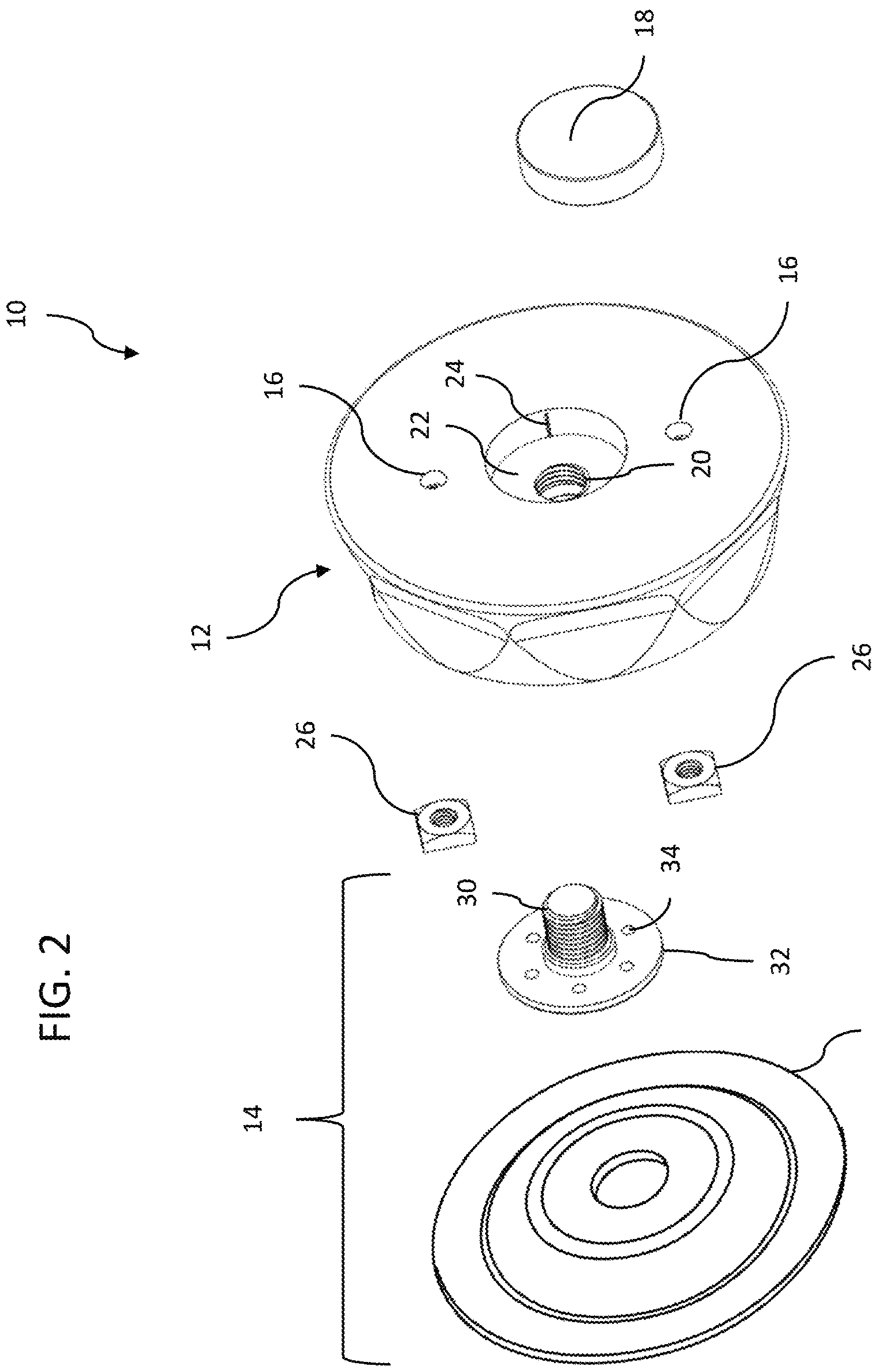


FIG. 2

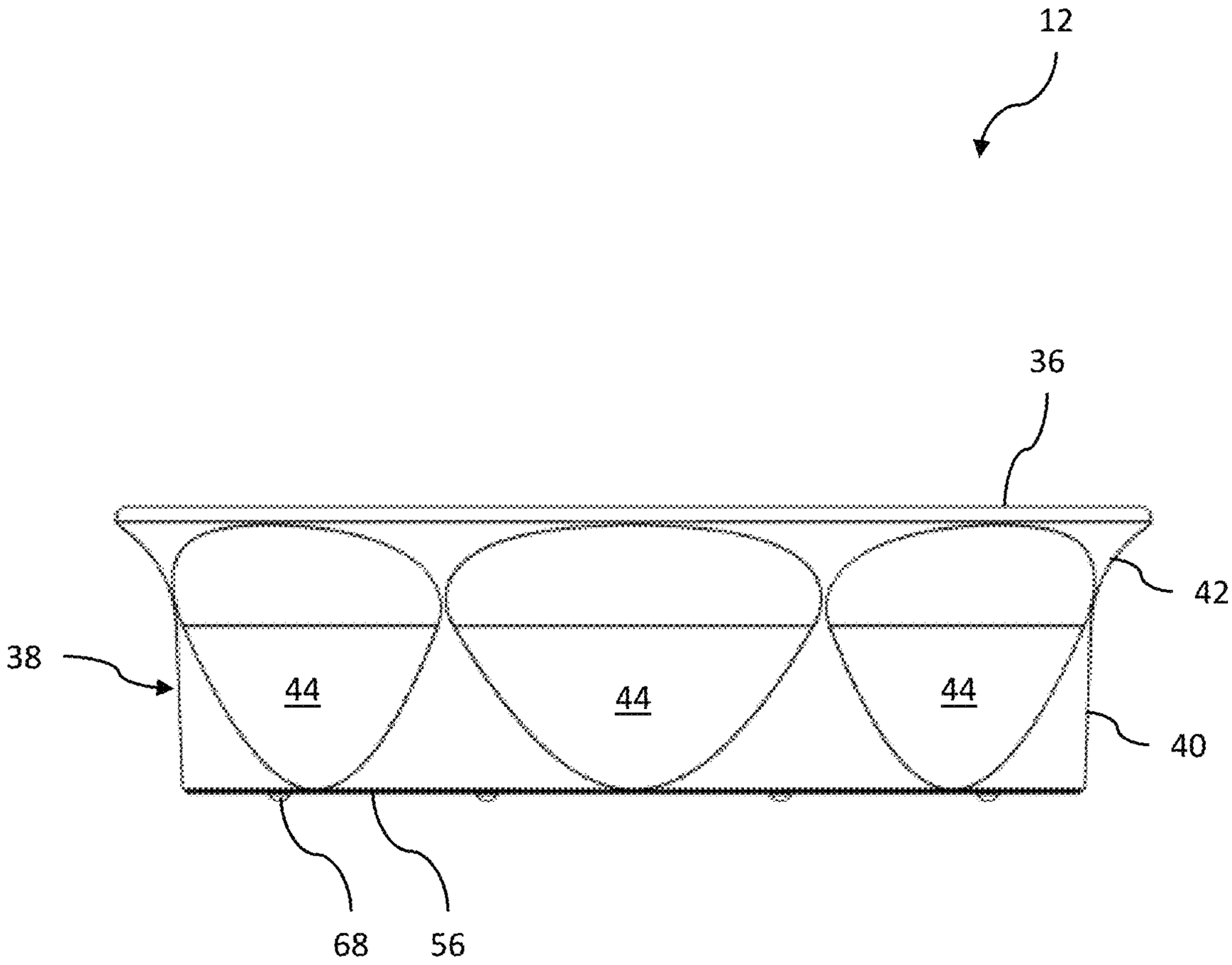


FIG. 3

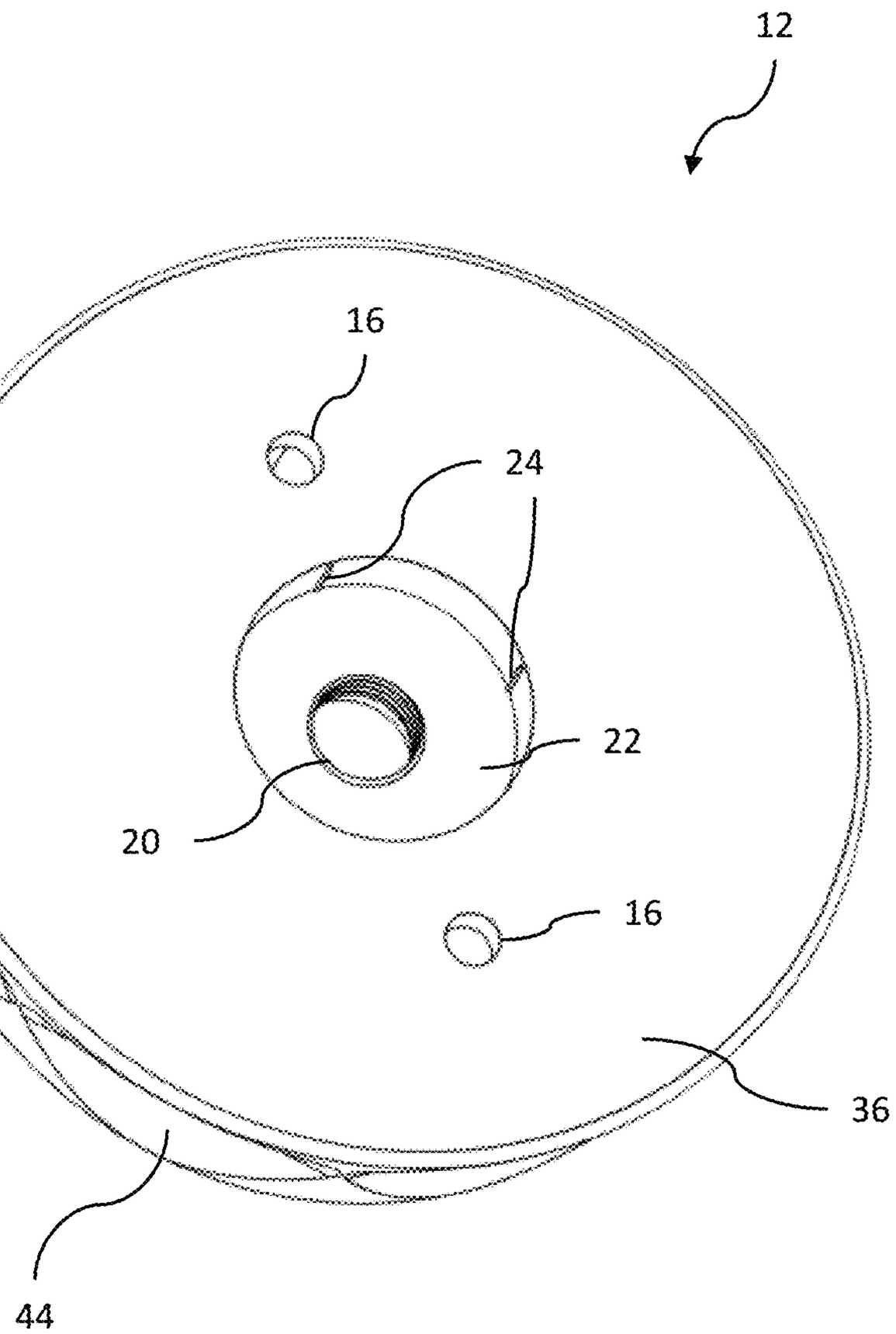


FIG. 4

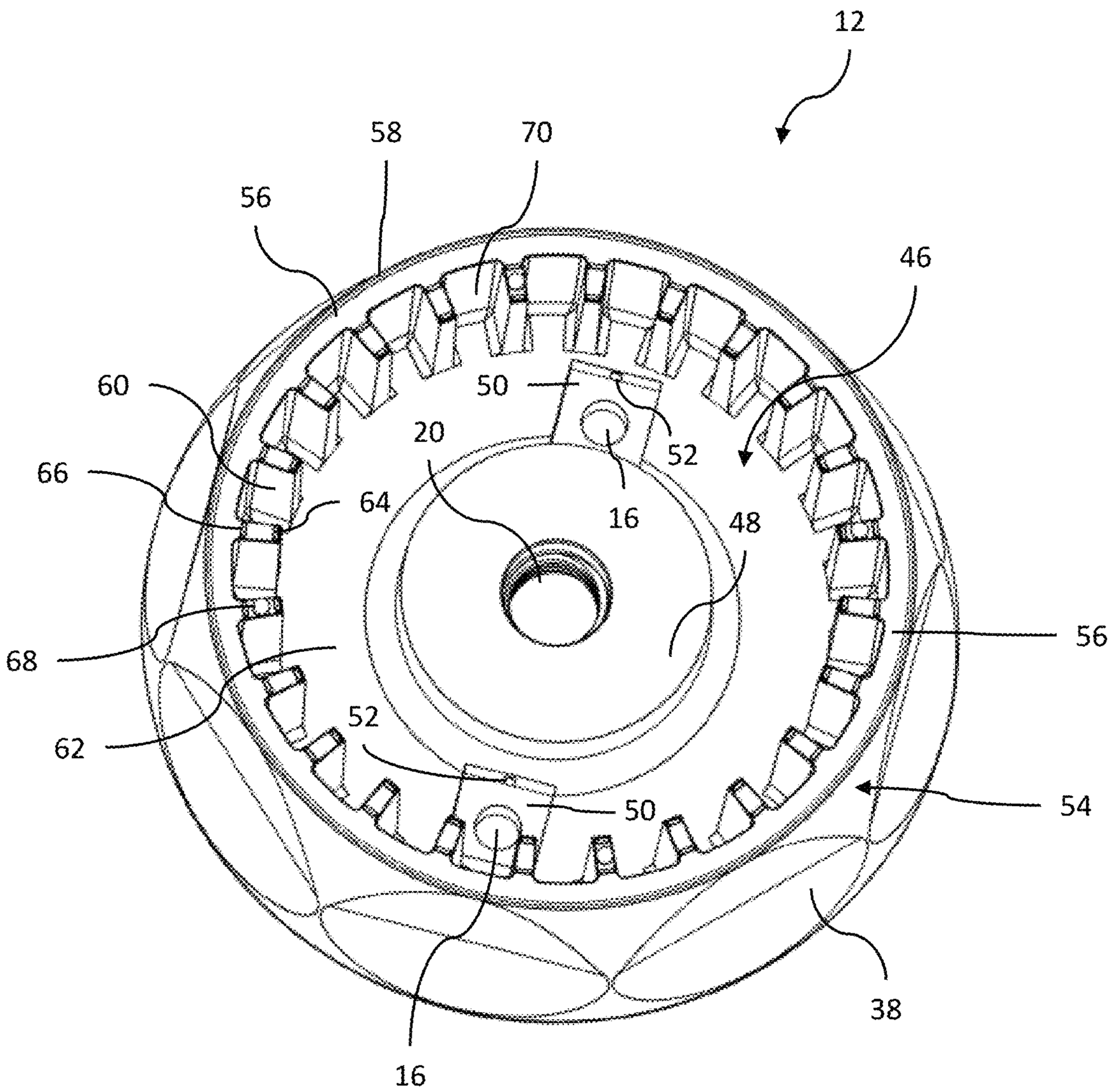


FIG. 5A

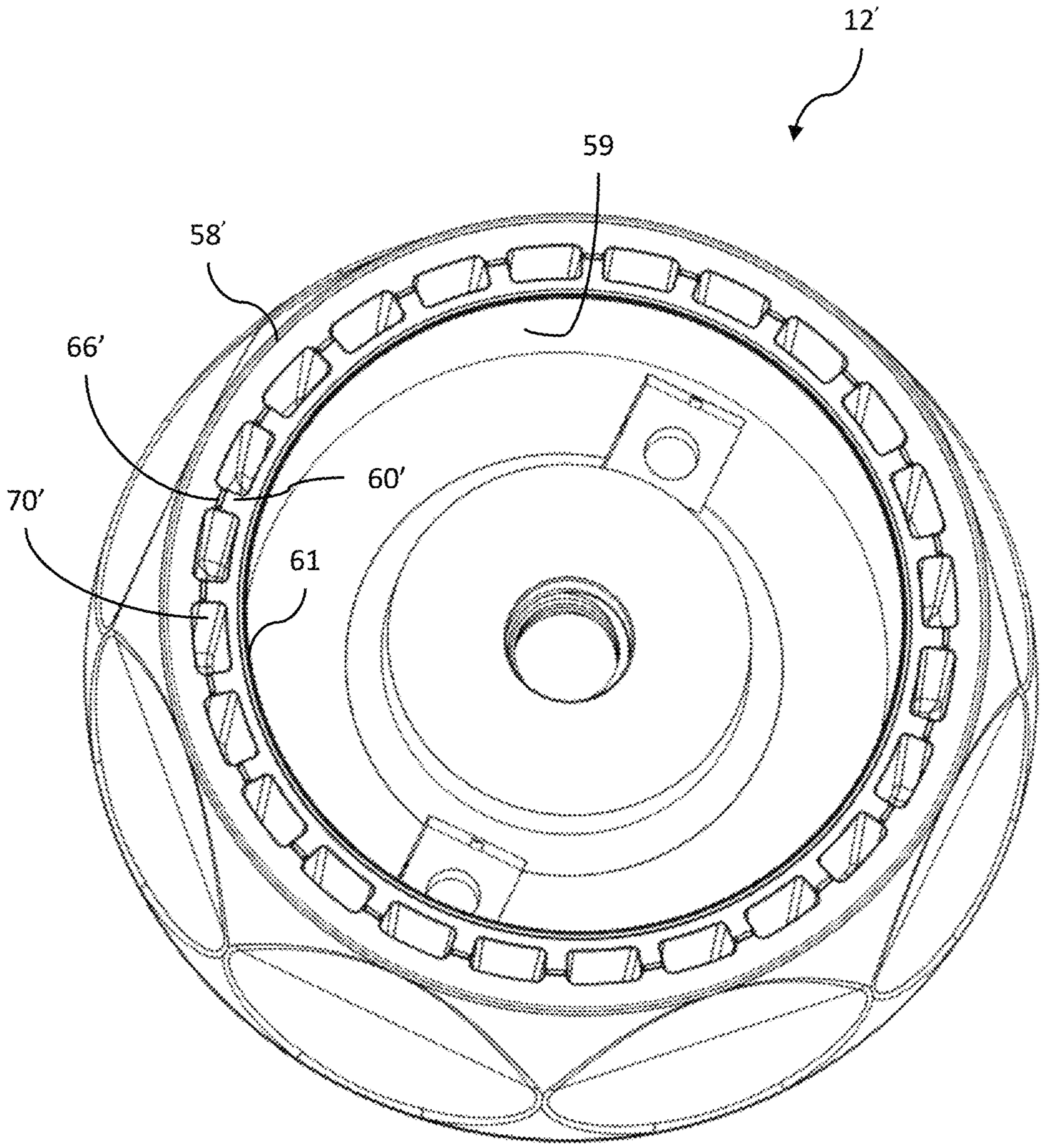


FIG. 5B

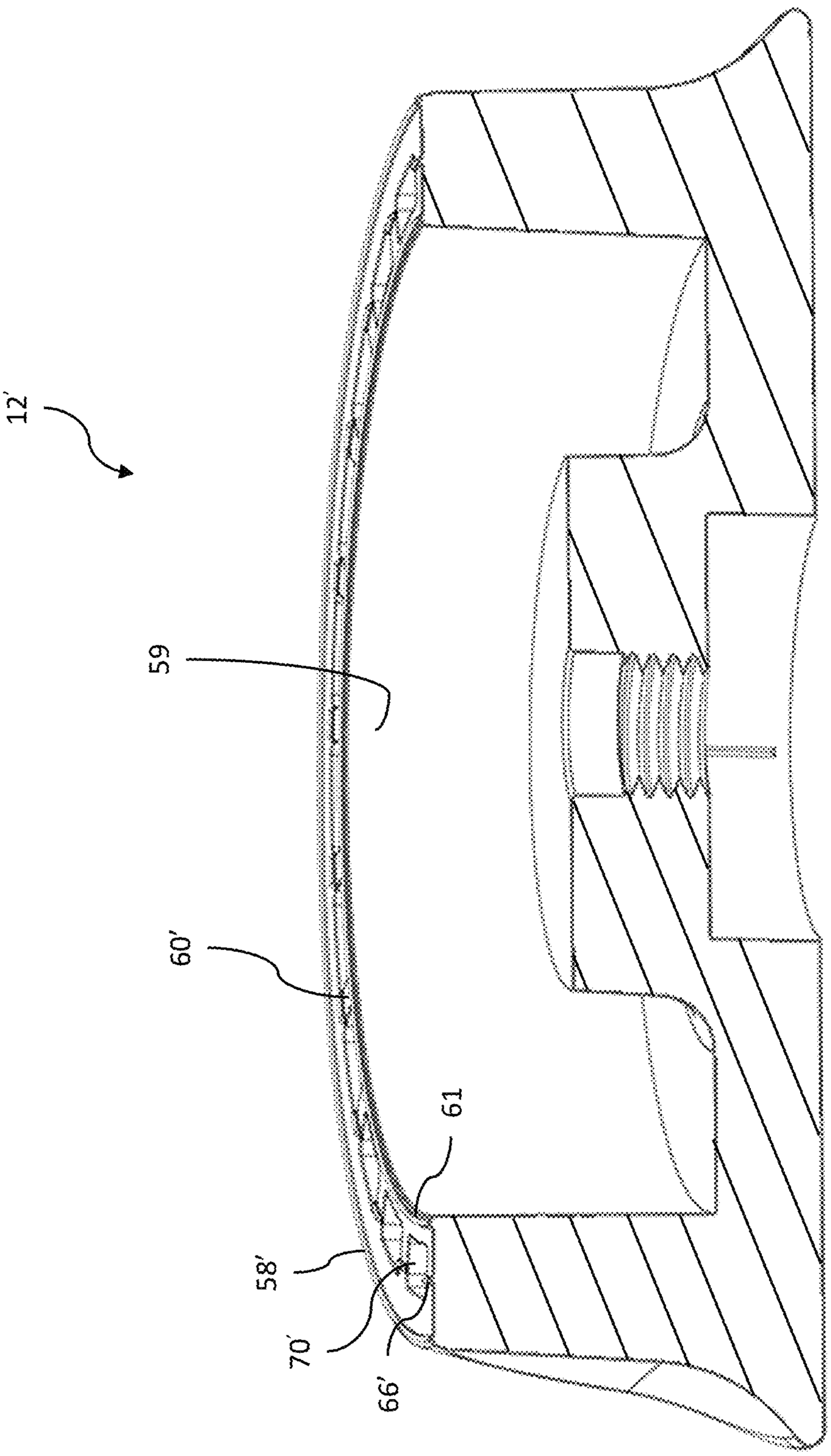


FIG. 5C

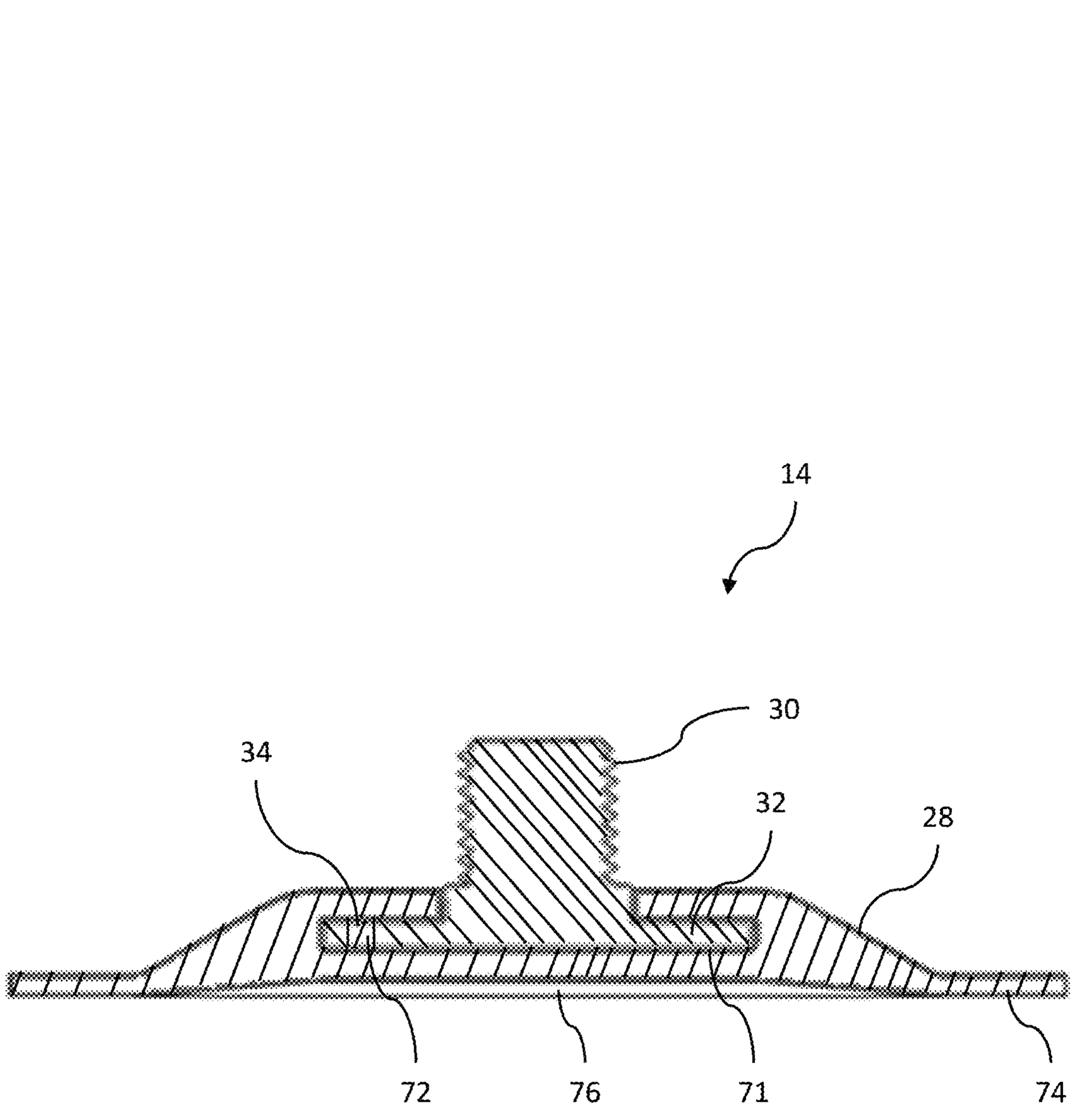


FIG. 6

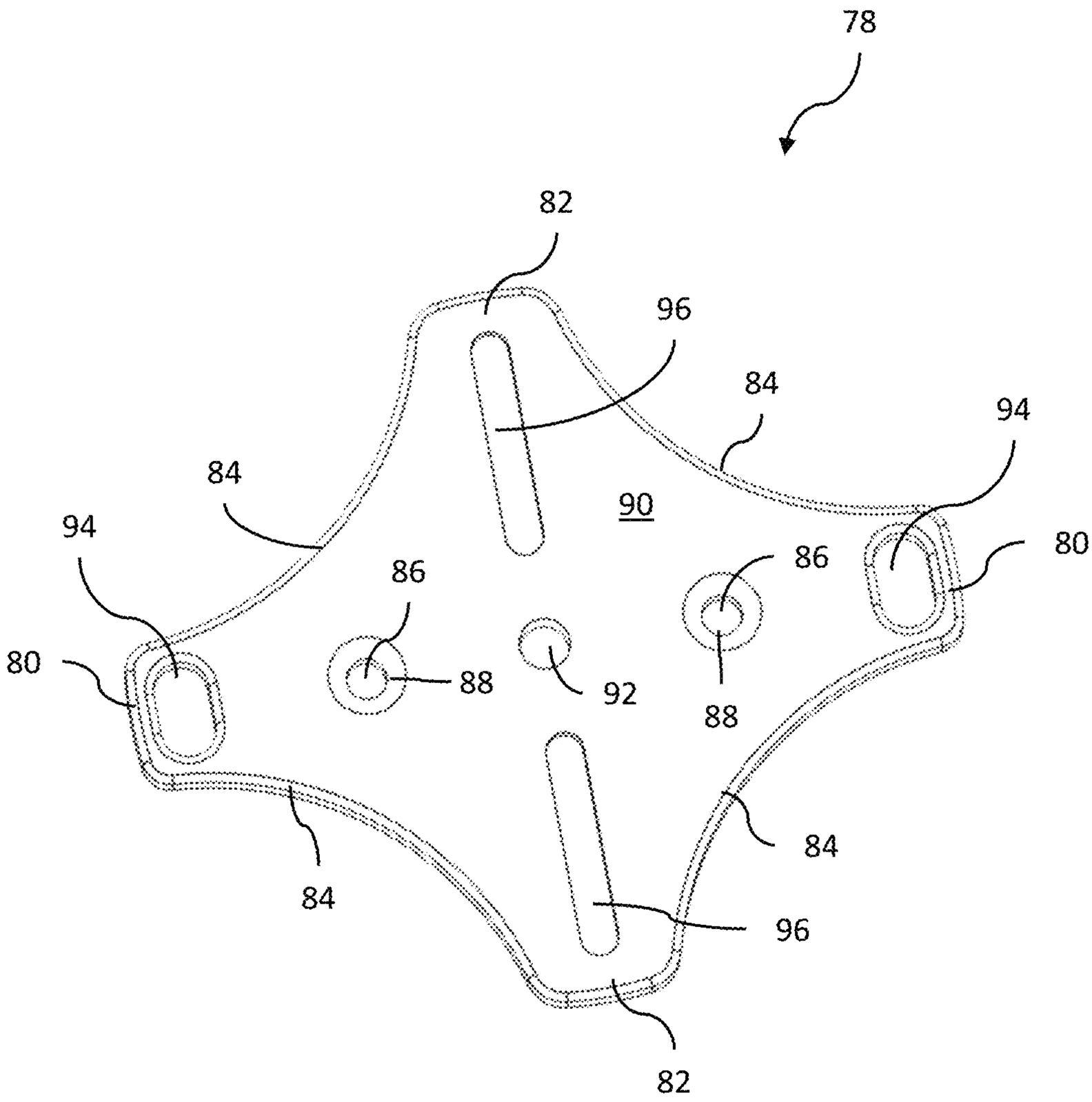


FIG. 7

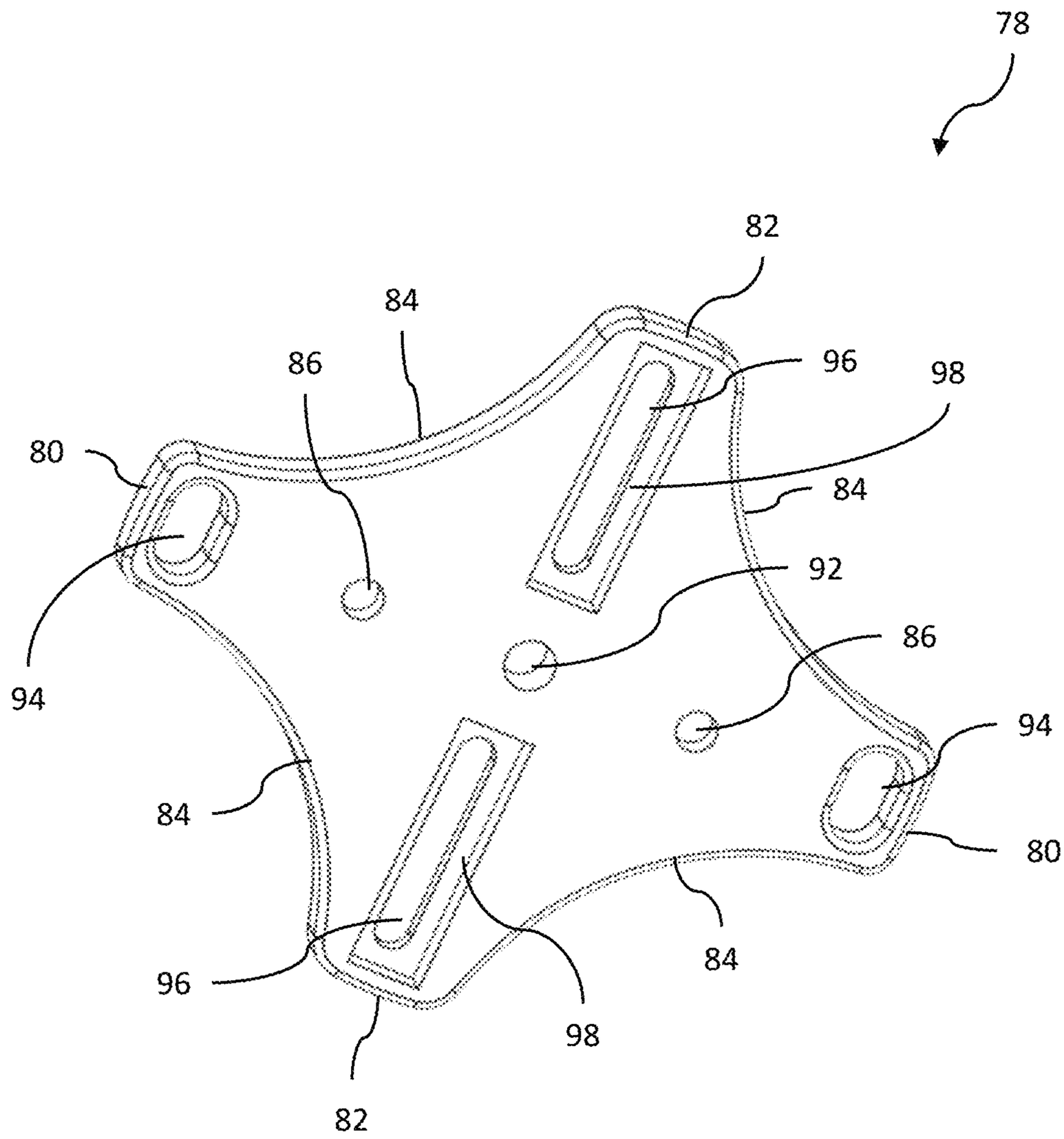


FIG. 8

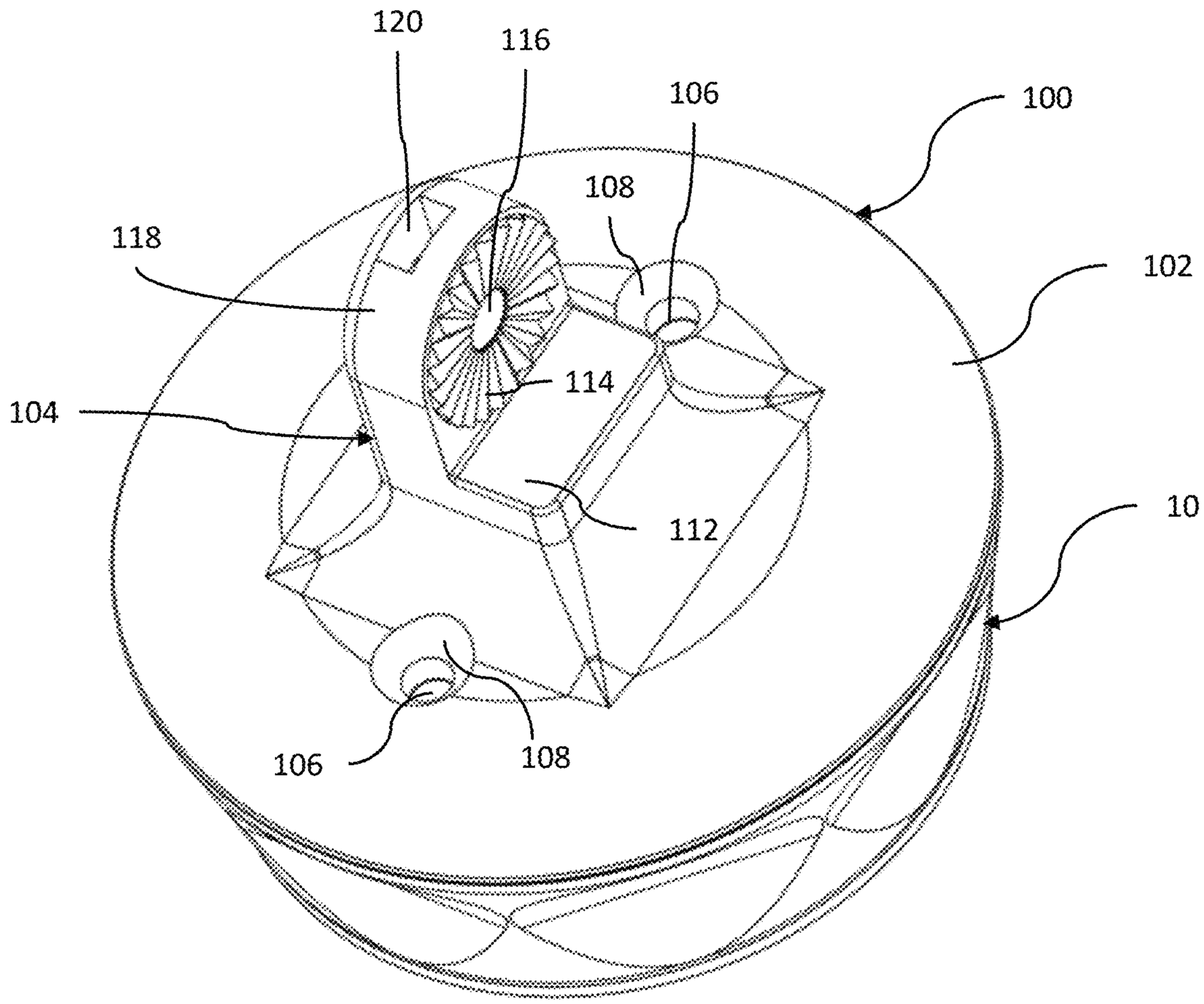


FIG. 9

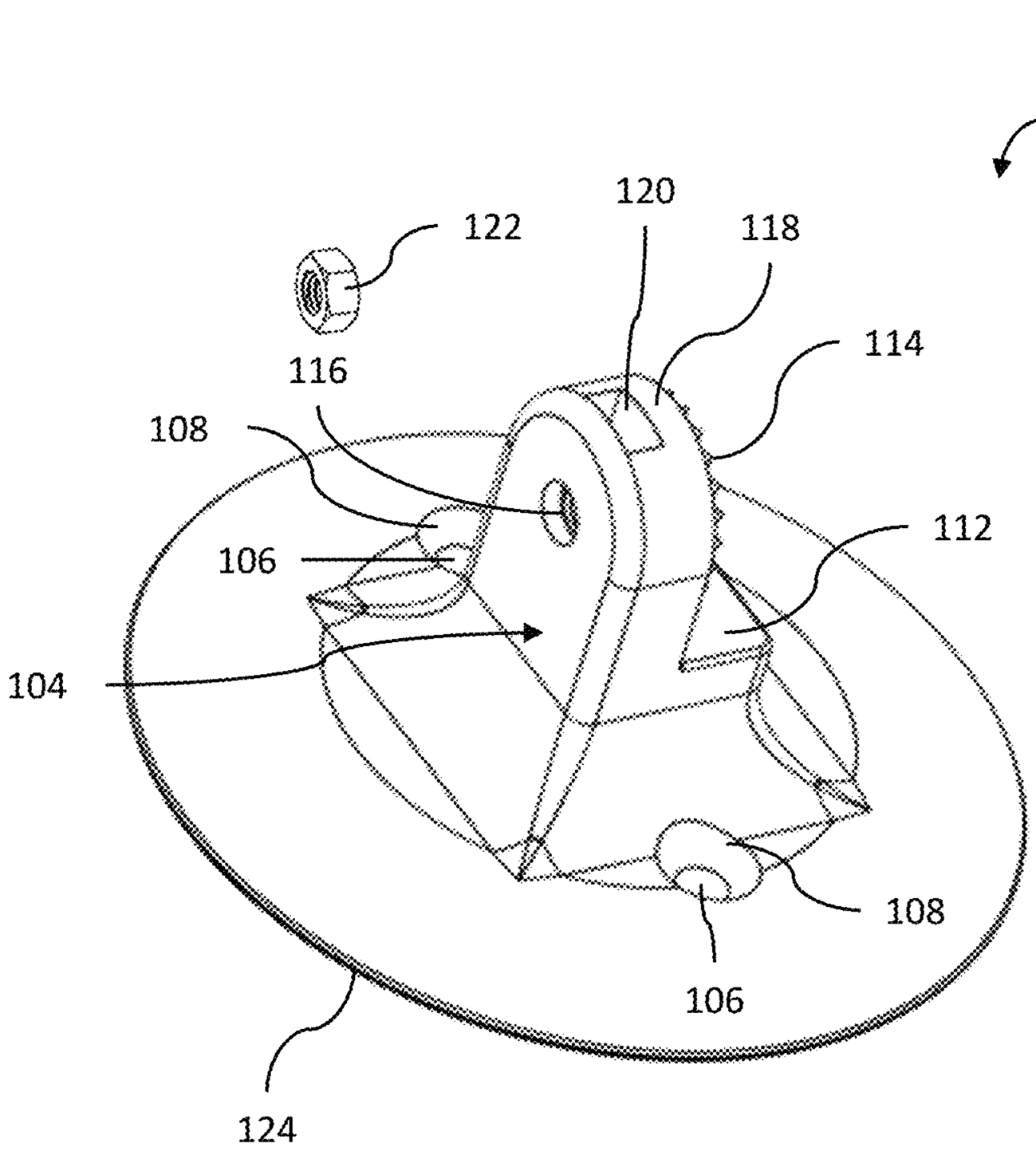


FIG. 10

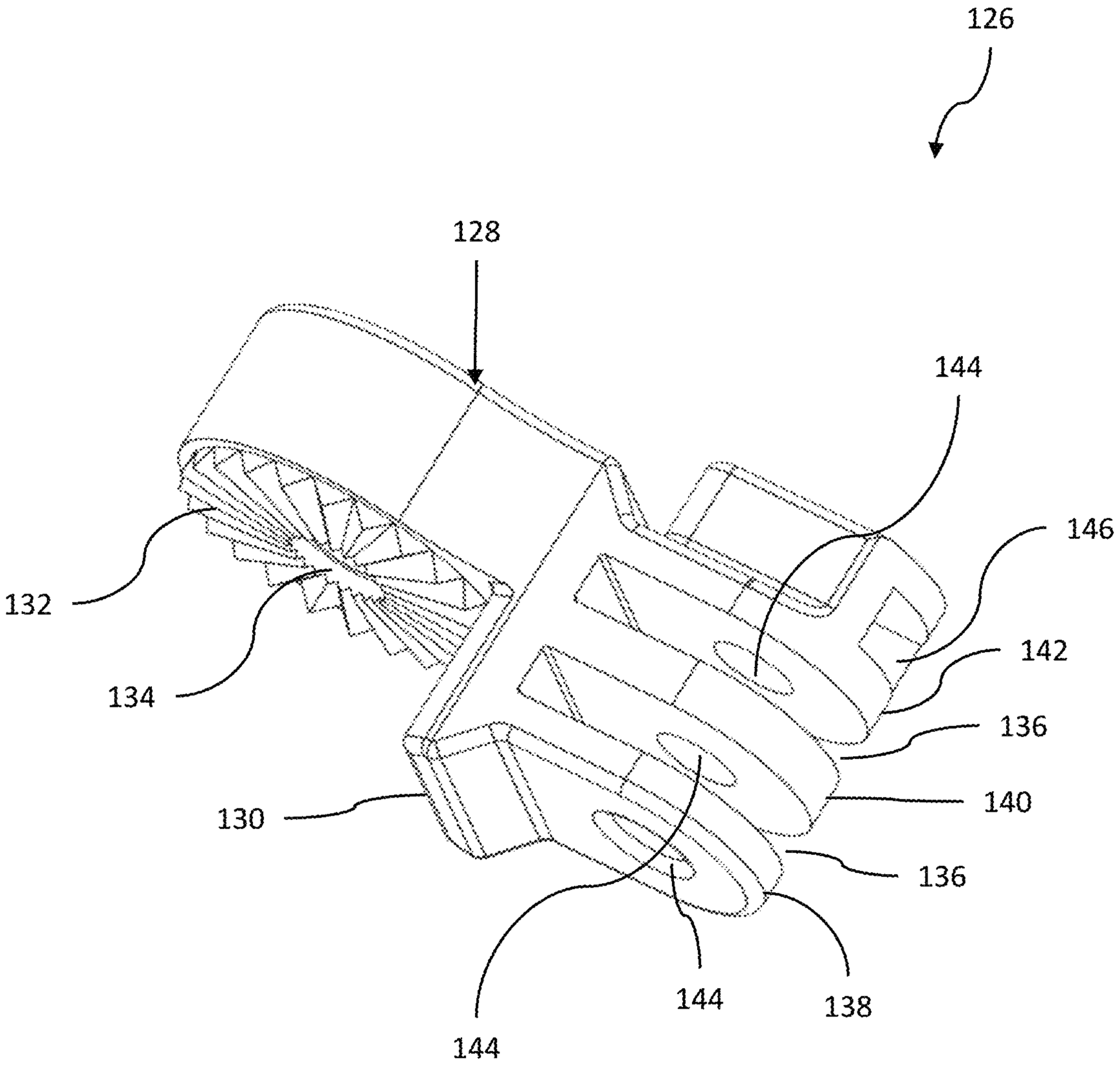


FIG. 11

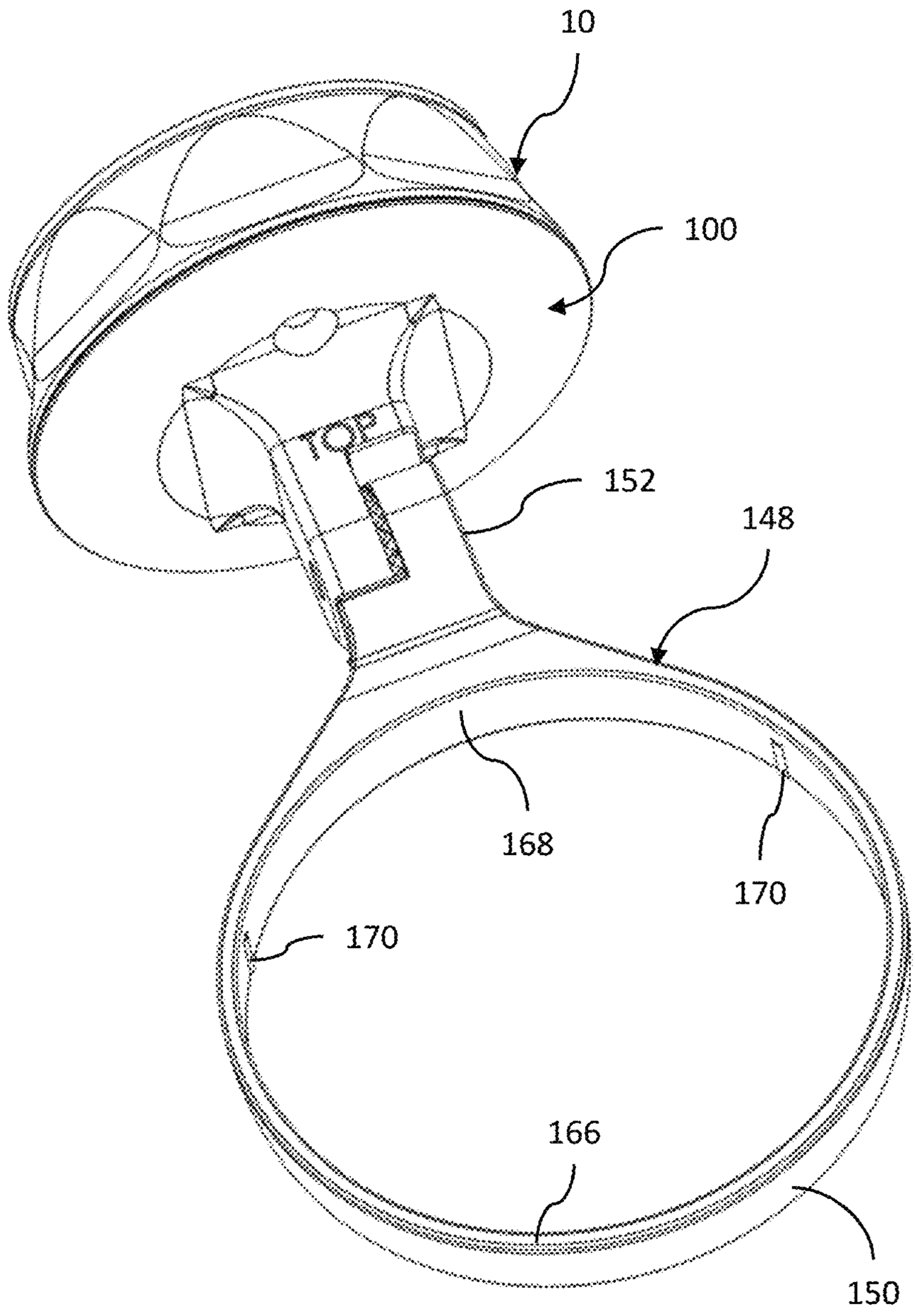


FIG. 12

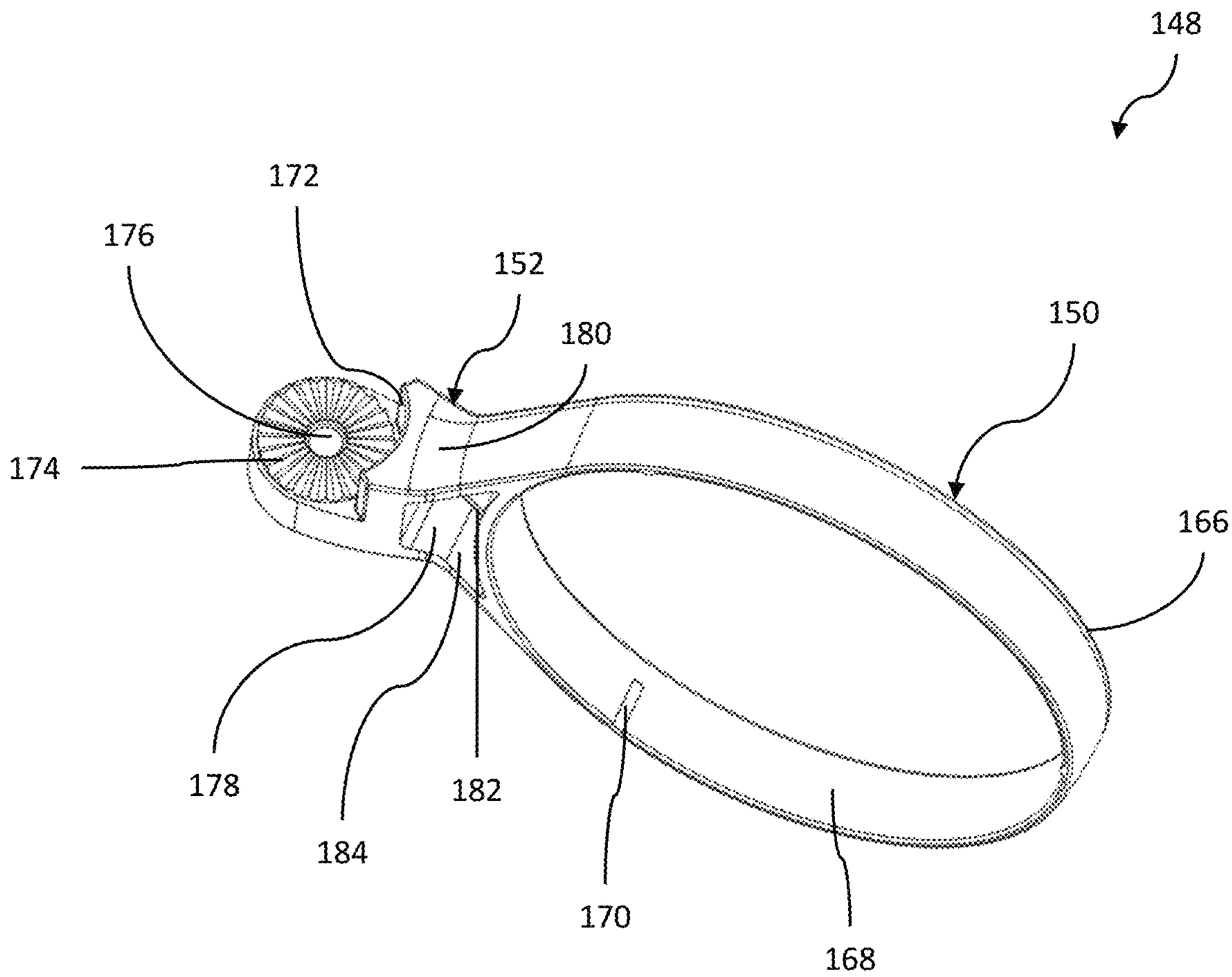


FIG. 13

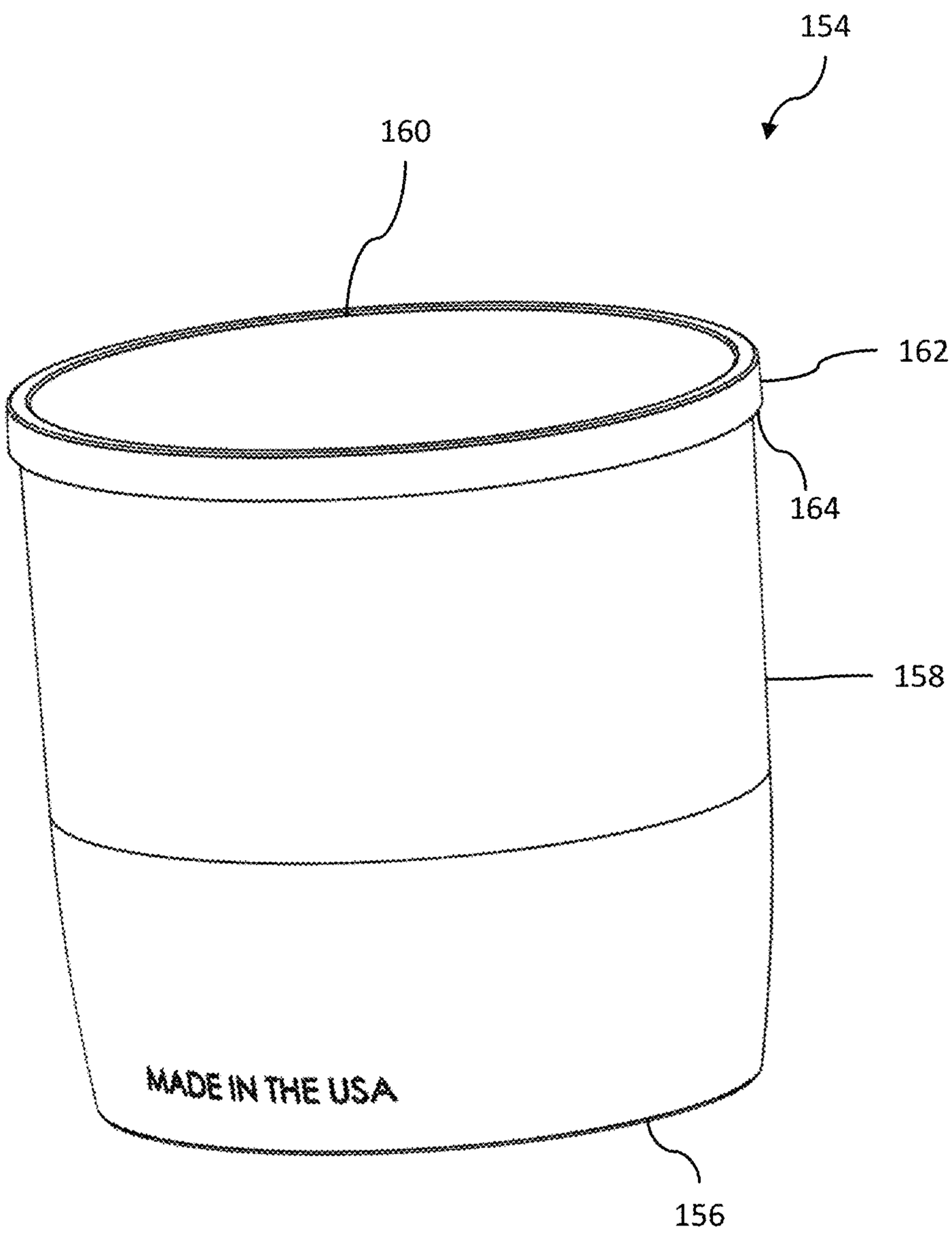


FIG. 14

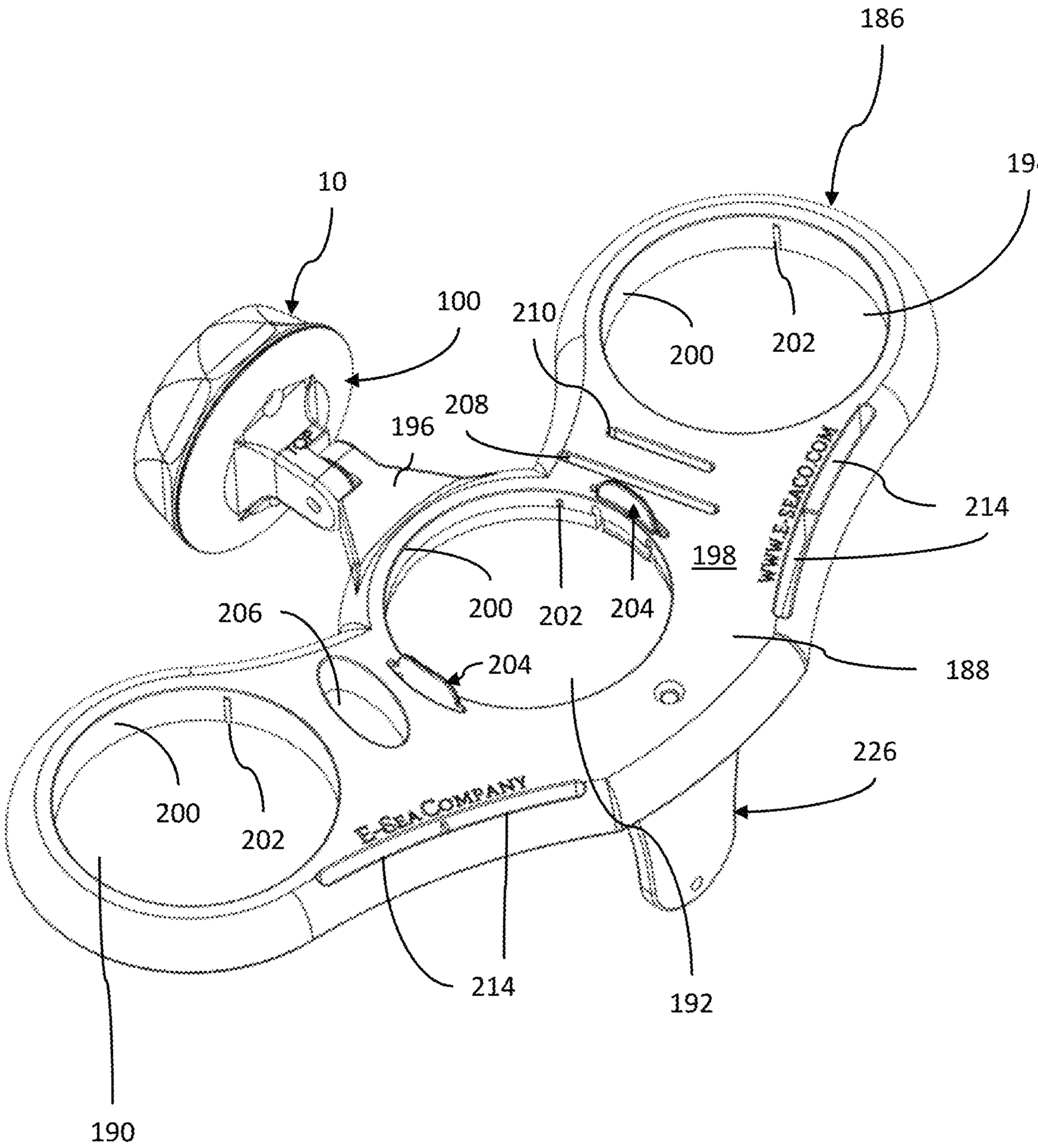

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FIG. 15

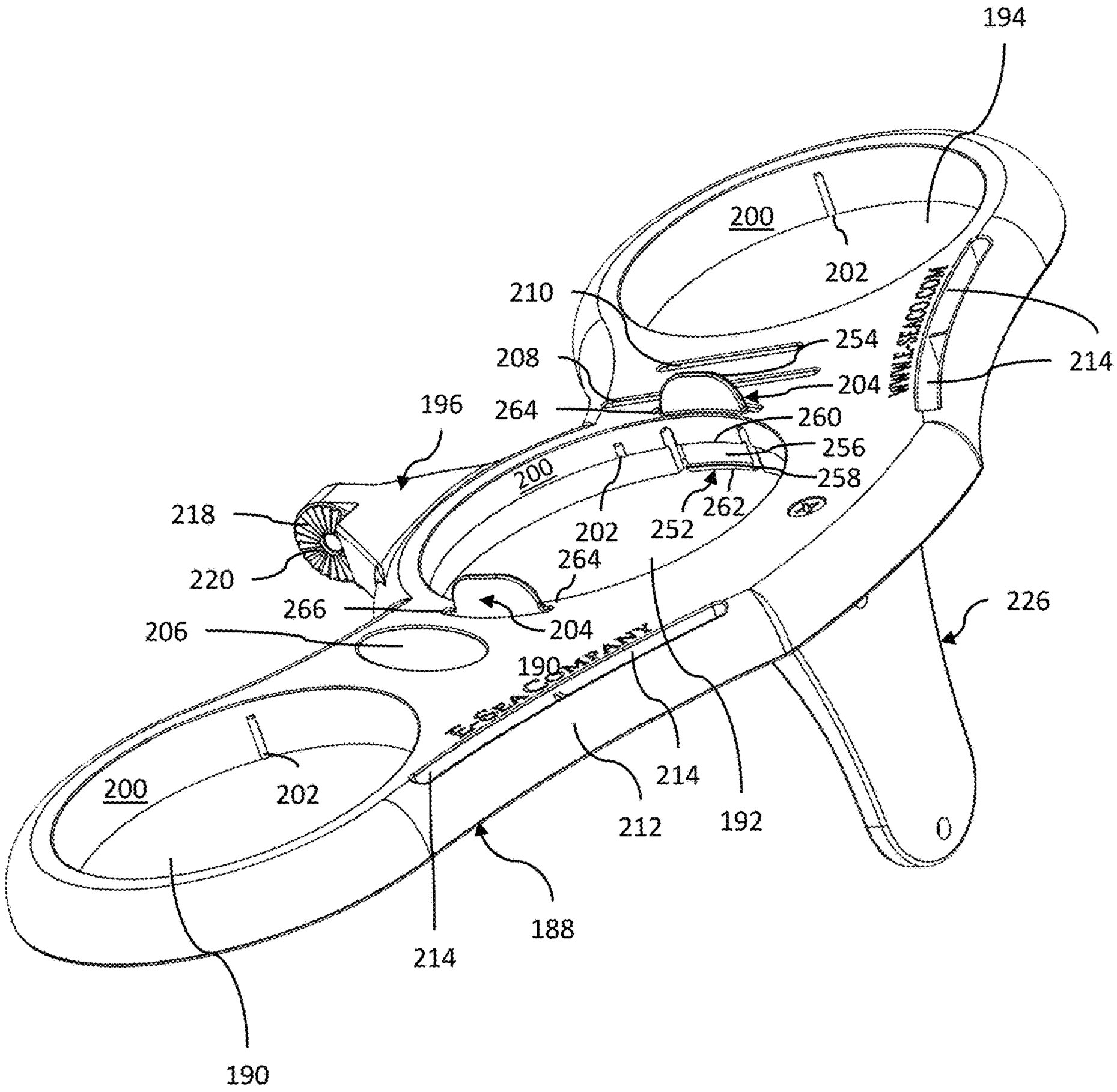

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FIG. 16

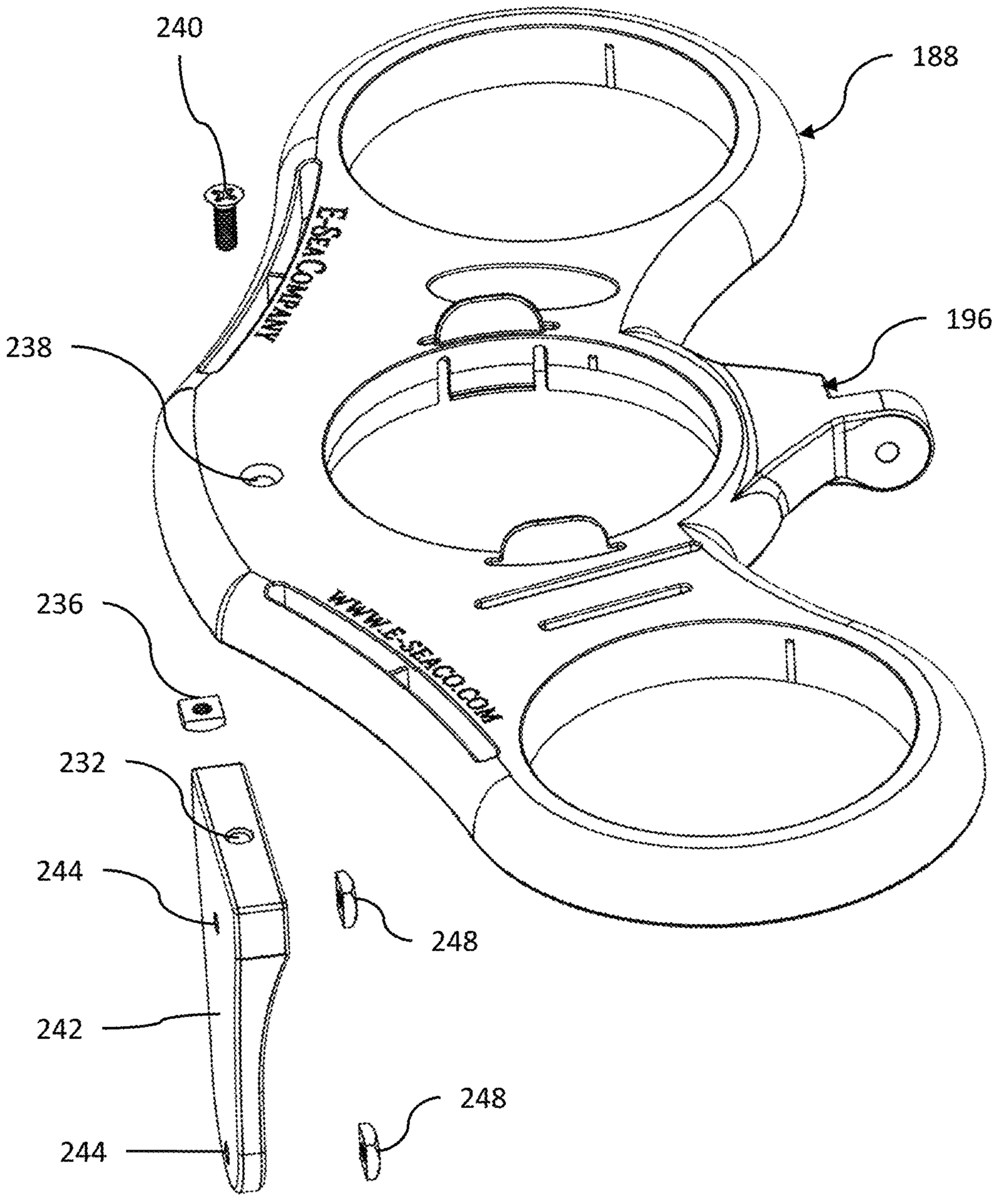


FIG. 17

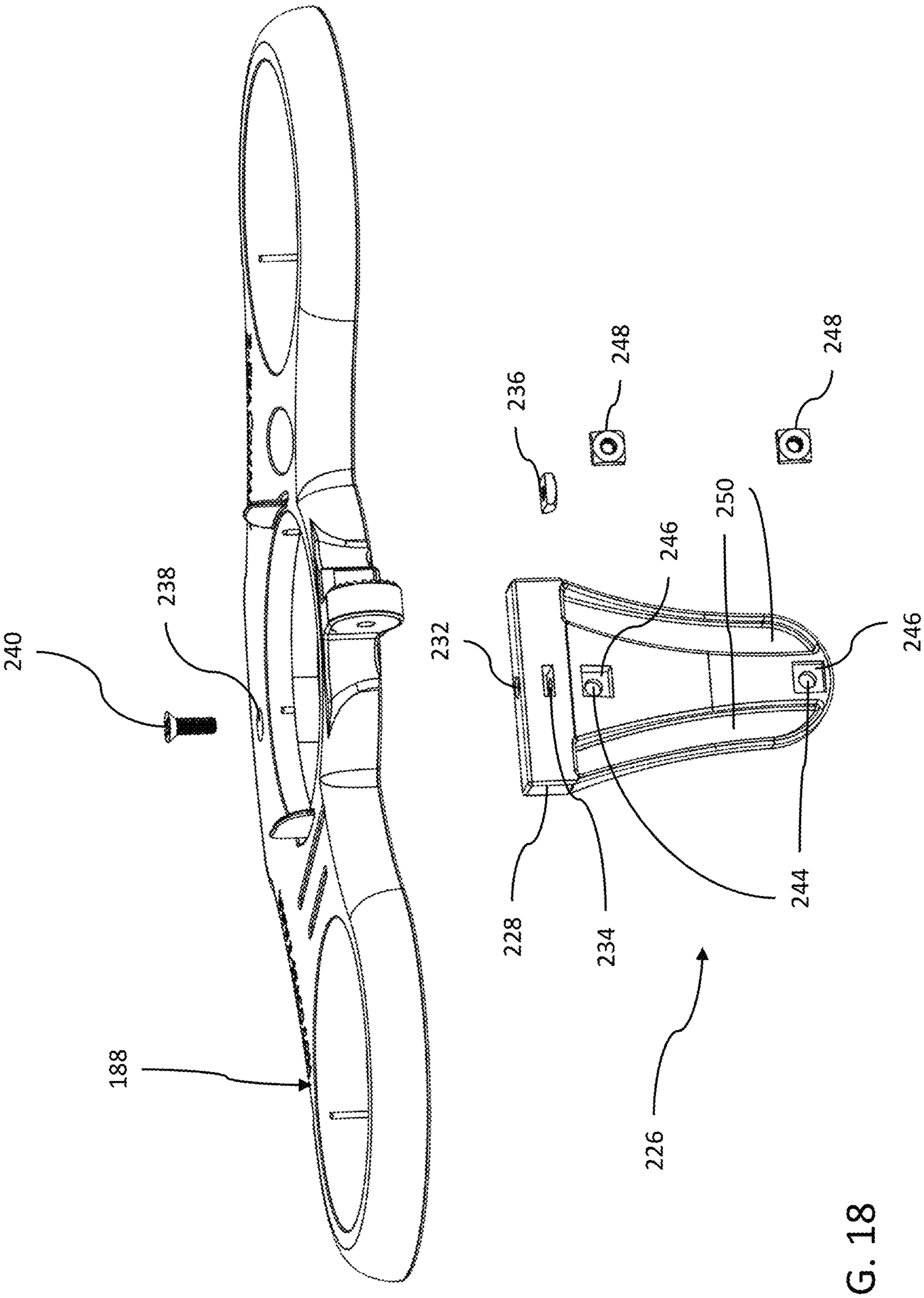


FIG. 18

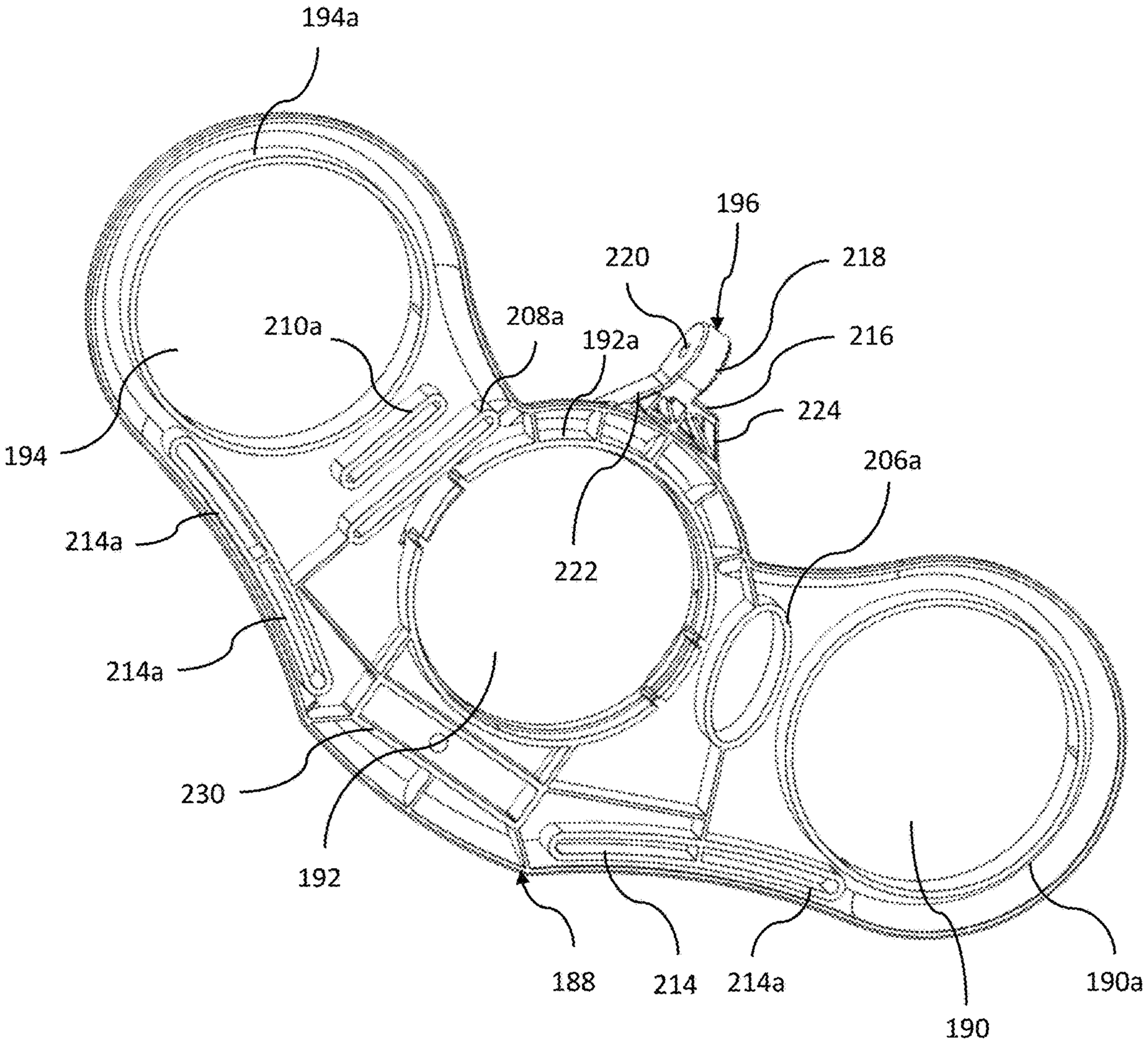

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FIG. 19

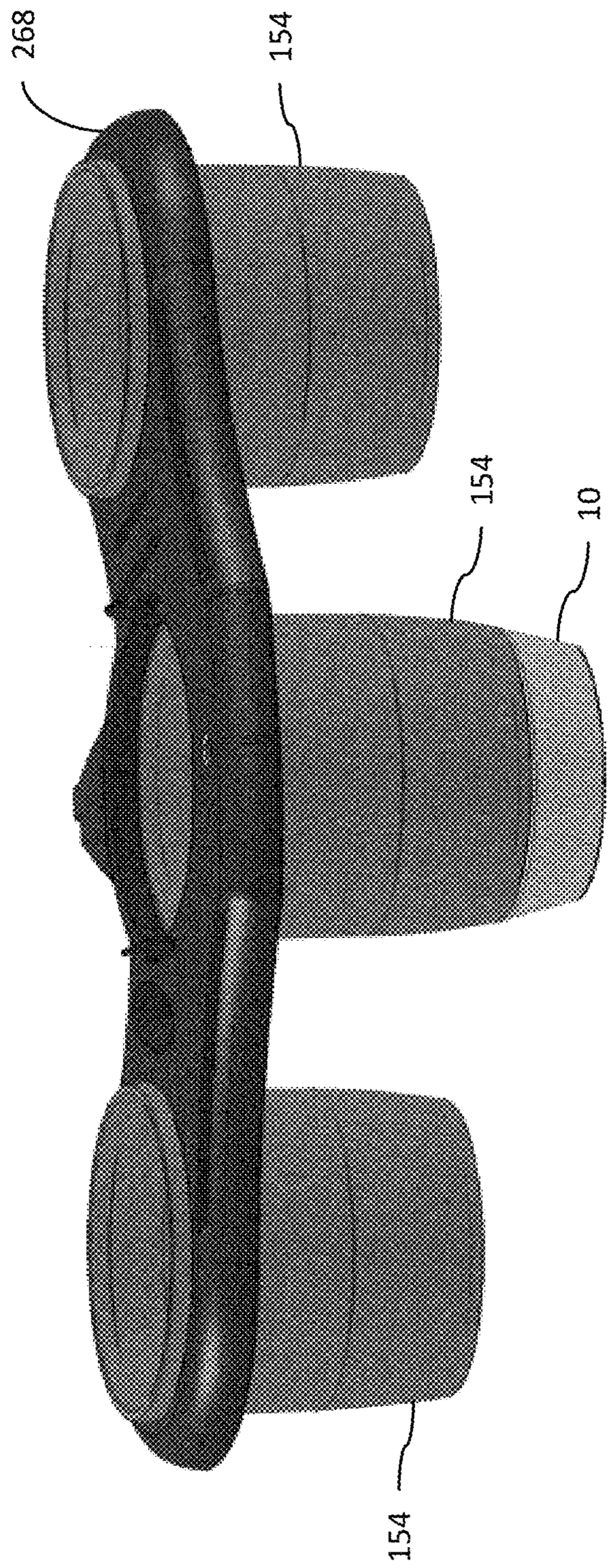


FIG. 20

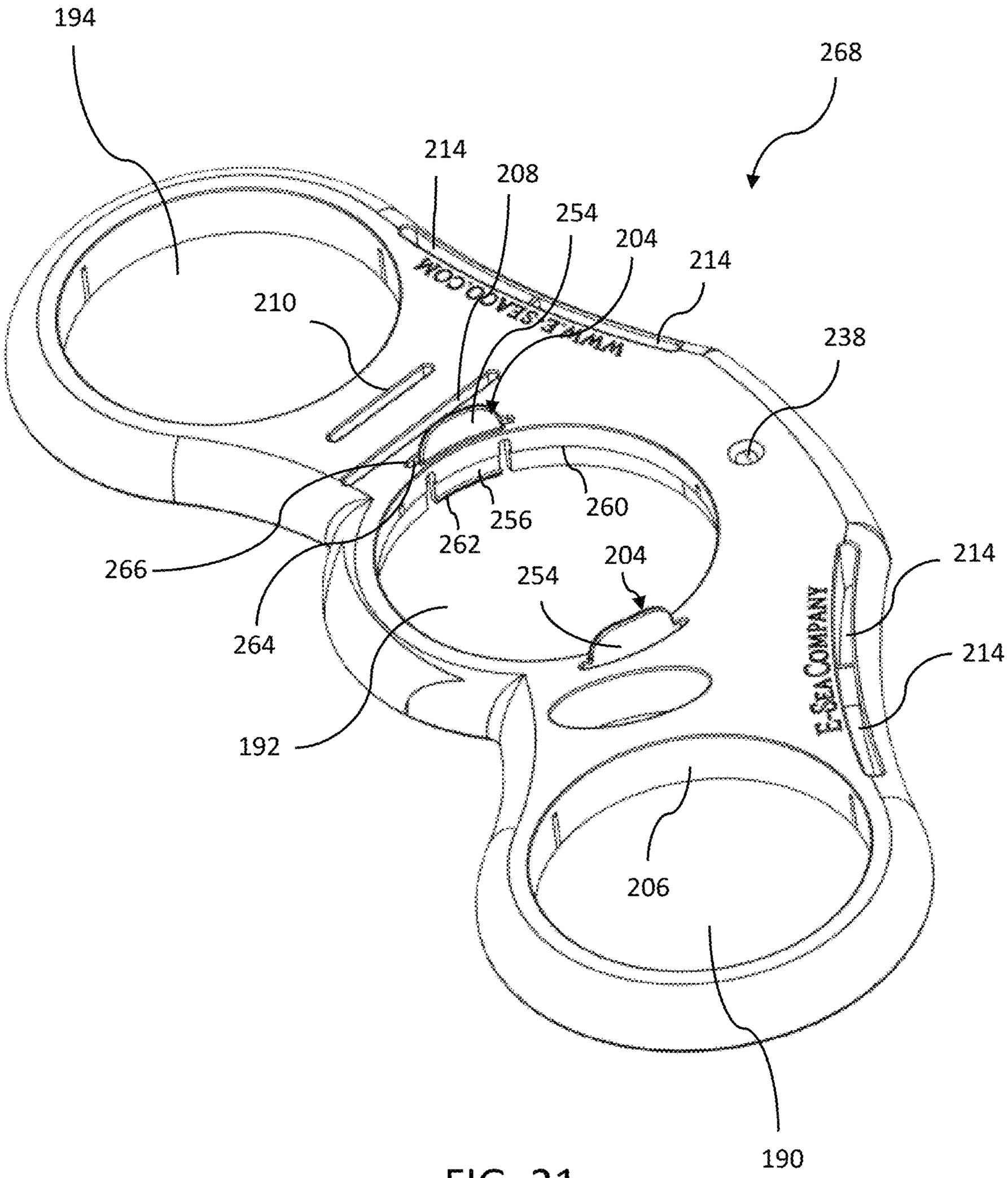

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FIG. 21

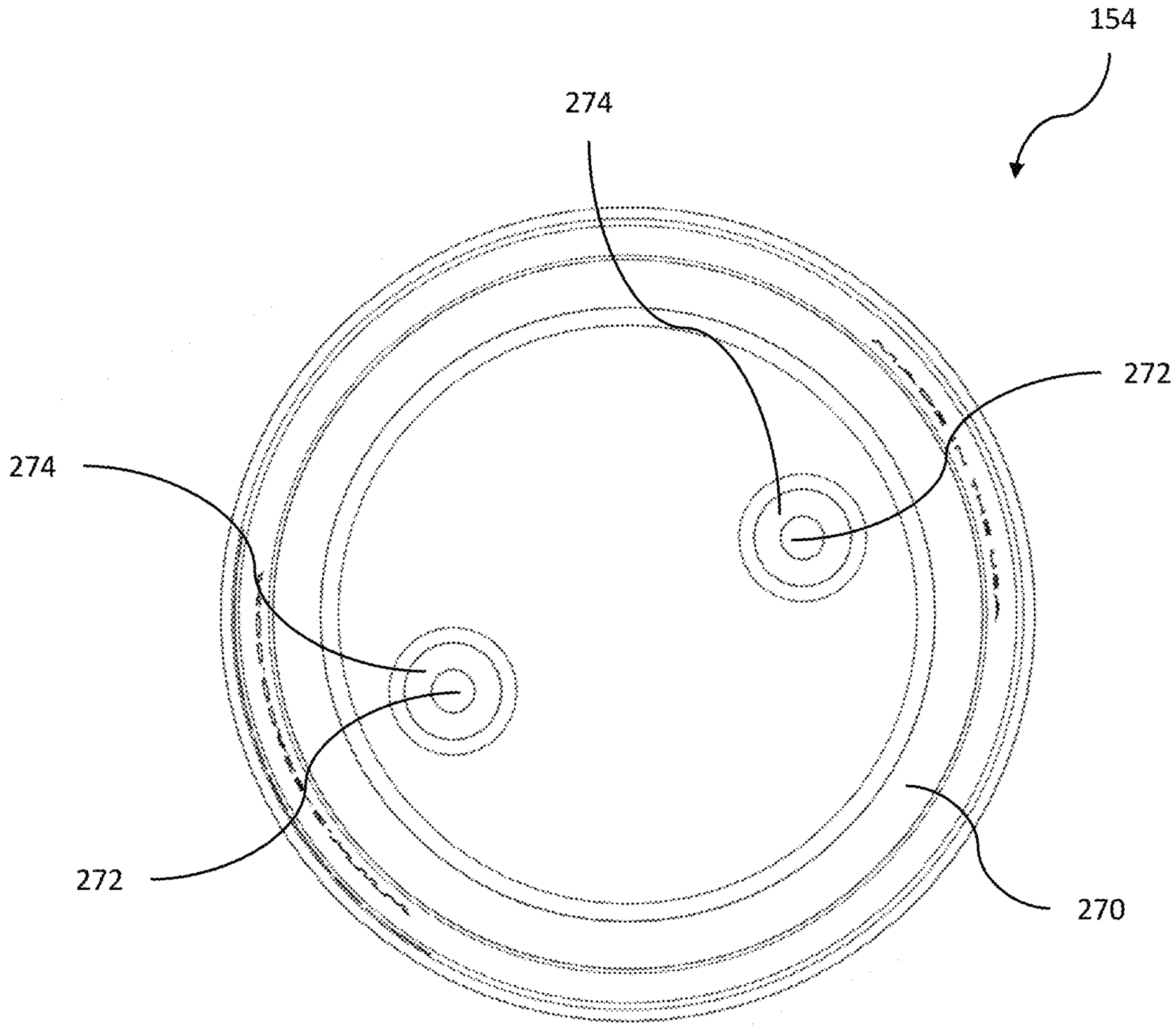


FIG. 22

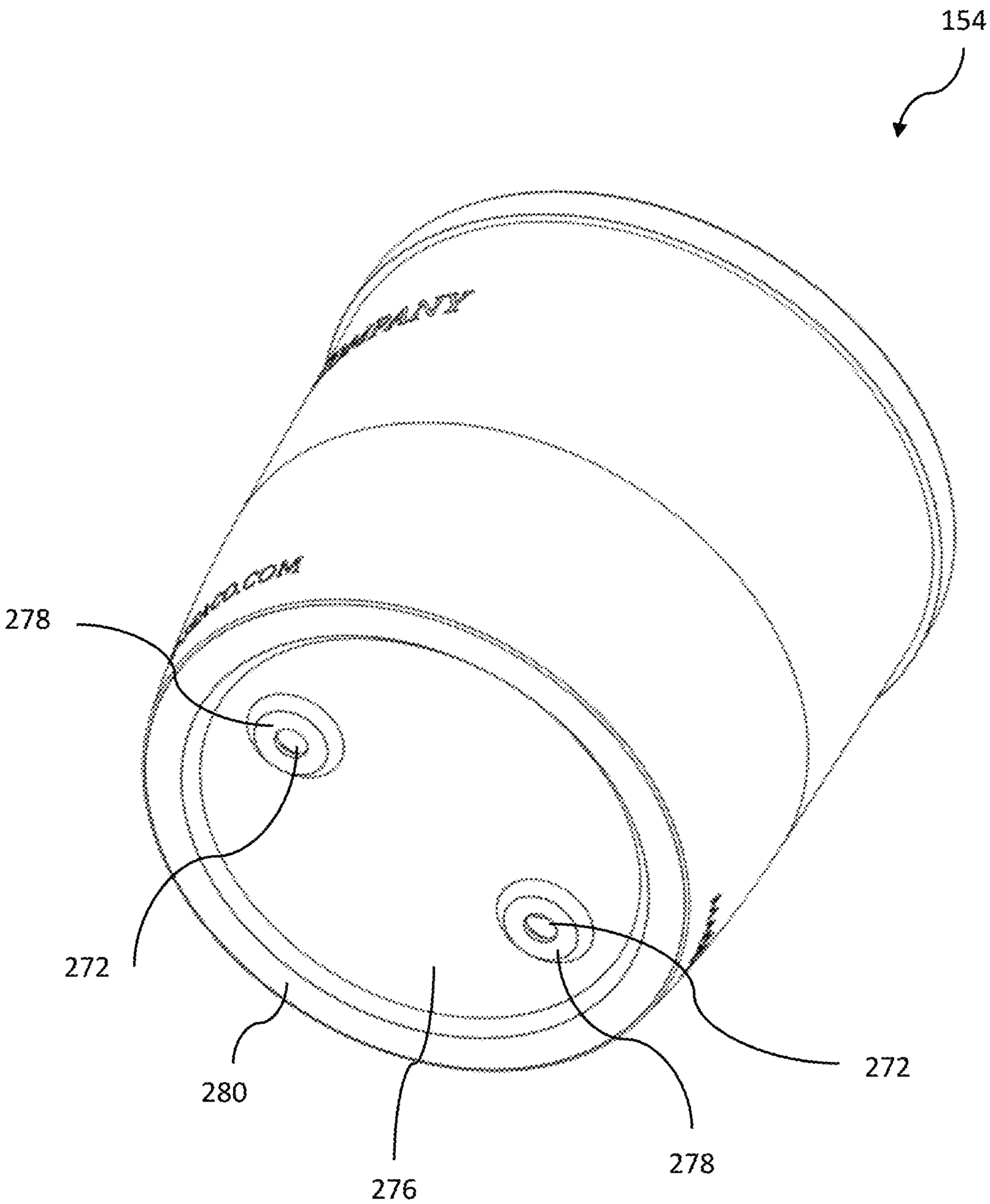


FIG. 23

SUCTION MOUNT AND BRACKETS AND ACCESSORIES THEREFOR

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. Provisional Application No. 62/959,424, filed Jan. 10, 2020, which is hereby incorporated by reference.

FIELD

This disclosure relates to suction cup mounts and, more particularly, to suction cup mounts affixed with a threaded operation and brackets and accessories for mounting to the suction cup mounts.

BACKGROUND

Suction cup mounts come in many shapes and sizes. There also are different methods for actuating suction mounts. For example, suction mounts using a vacuum pump include an air pump that must be manually pressed by the user to attach to a surface. The air pump makes vacuum pump suction mounts relatively bulky and more difficult to attach to surfaces.

Another method to attach suction cup mounts is a lever activated system where the lever or switch pulls the center of the suction cup upward, creating the necessary vacuum between the surface and the suction cup. Lever activated suction mounts require operating space directly above and around the mount in order to operate the lever. Due to the location of the lever or switch, attaching an accessory directly above or around the mount is generally not realistic.

A further method includes a suction cup mount activated using a threaded connection with the suction cup. In a common form, there is a threaded bolt, a cover plate and a threaded nut. The nut is tightened to pull the center of the suction cup upward and locks accessories in between the suction cup and nut. While this style of suction mount can be very compact, there often are many pieces involved and a very limited number of accessories can be attached to it. Indeed, many times the mount is dedicated to a specific accessory.

Thus, there is a need for a suction mount that easily creates enhanced suction, has a relatively compact design and supports different accessories.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of a suction mount;

FIG. **2** is an exploded view of the suction mount of FIG. **1**;

FIG. **3** is a side elevation view of the mount body of the suction mount of FIG. **1**;

FIG. **4** is a top perspective view of the mount body of the suction mount of FIG. **1**;

FIG. **5**A is a bottom perspective view of the mount body of the suction mount of FIG. **1**;

FIG. **5**B is a bottom perspective view of another mount body;

FIG. **5**C is a cross-section view of the mount body of FIG. **5**B;

FIG. **6** is a cross-section view of a suction cup assembly of the suction mount of FIG. **1**.

FIG. **7** is a top perspective view of a bracket used to mount accessories to the suction mount of FIG. **1**;

FIG. **8** is a bottom perspective view of the bracket of FIG. **7**;

FIG. **9** is top perspective view of an adjustable angle mount attached to the suction mount of FIG. **1**;

FIG. **10** is an exploded view of the adjustable angle mount of FIG. **9**;

FIG. **11** is a perspective view of a double slotted accessory bracket;

FIG. **12** is a perspective view of a single cup holder accessory attached to the suction mount of FIG. **1** using the adjustable angle mount of FIG. **10**;

FIG. **13** is a perspective view of the single cup holder accessory of FIG. **12**;

FIG. **14** is a perspective view of a cup accessory;

FIG. **15** is a perspective view of a multi-cup accessory mounted to the suction cup of FIG. **1** using the adjustable mount of FIG. **9**;

FIG. **16** is a perspective view of the multi-cup accessory of FIG. **15**;

FIG. **17** is an exploded view of the multi-cup accessory of FIG. **16**;

FIG. **18** is another exploded view of the multi-cup accessory of FIG. **16**;

FIG. **19** is a bottom perspective view of the multi-cup accessory of FIG. **16**;

FIG. **20** is a perspective view of the multi-cup accessory using the cup accessory of FIG. **14** to mount directly to the suction mount of FIG. **1**;

FIG. **21** is a perspective view of the multi-cup accessory of FIG. **20**;

FIG. **22** is a plan view of the cup accessory of FIG. **14**; and

FIG. **23** is a bottom perspective view of the cup accessory of FIG. **14**.

DETAILED DESCRIPTION

With reference to FIG. **1**, there is illustrated a suction mount **10** having a mount body **12** and a suction cup assembly **14**. The mount body **12** and the suction cup assembly **14** are mechanically interconnected by a threaded post **30**. The mount body **12** is easily turned to increase the suction force of the suction cup assembly **14** and to maintain the suction cup assembly in place. The mount body **12** includes a pair of mounting holes **16** used to attach accessories and a universal mounting bracket, as described further below. The mount body **12** also can include a magnet **18** used to attach magnetic objects. As seen, the suction mount **10** has a compact body that can be used to mount a variety of accessories. It also can be easily scaled between a relatively large size for large accessories to a small, compact and slim size for small accessories. Regardless of the size, the mount body **12** has few moving parts and can provide a low center of gravity.

As shown in FIG. **2**, the suction mount **10** includes a centrally located threaded hole **20** that cooperates with the suction cup **14**. Overlying and centered on the threaded hole **20** is a recess **22** for the magnet **18**. The recess **22** includes small longitudinally extending ribs **24** to help frictionally engage the magnet **18** to maintain the magnet **18** in the recess **22** with a friction fit. The magnet **18** also may be secured in the recess **22** with an adhesive. A pair of nuts **26** are mounted to an inside of the mount body **12** at the accessory mount holes **16** to cooperate with screws of the desired bracket or accessory. The suction cup assembly **14** includes a suction cup **28** and the threaded post **30** with a base **32**. The base **32** is affixed to the suction cup **28** in any

manner, preferably, however, by overmolding. The base **32** includes holes **34** that enable material of the suction cup **28** to flow into during the molding process to secure the suction cup **28** to the base **32** so that the base **32** is not able to rotate relative to the suction cup **28**.

With reference to FIGS. **3-5**, the mount body **12** has an annular puck like configuration with a flat top surface **36**. An annular side surface **38** has a generally cylindrical portion **40** and outward curved portion **42** adjacent the top surface **36**. The side surface **38** includes concave recesses **44**. The outward curved portion **42** increases the size of the top surface **36** and coupled with the recesses **44** provides enhanced gripping for a user to turn the mount body **12** during mounting of the suction mount **10** to a surface.

The mount body **12** defines a generally hollow region **46** open at its underside. A boss **48** is centrally located in the hollow region **46** and defines at least in part the threaded hole **20**. The mount body **12** defines two pockets **50** located diametrically opposite one another about the boss **48**. Each pocket **50** is sized to receive one of the complementary sized nuts **26** in a manner that retains the nut **26** with a friction fit. The nuts **26** align with the accessory mount holes **16**. Each pocket **50** may include ribs **52** to further enhance the friction fit between the nut **26** and the pocket **50**. The nuts **26** may be overmolded into the pockets **50**.

An annular wall **54** surrounds the hollow region **46**. The annular wall **54** includes the outer annular side surface **38** and an annular end surface **56**. The annular end surface **56** includes an annular rib **58** extending from its outer circumference. A series of longitudinal ribs **60** extend radially inward from the annular wall **54** and from a base **62** of the hollow region **46**. A terminal end of each rib **60** includes an inner rib **64** and an outer rib **66**. The terminal end of four of the longitudinal ribs **60** include a pad **68**. The pads **68** can be located on ribs 90 degrees apart and extend further than the ribs **60**, **64**, **66** (FIG. **3**). The annular rib **58**, inner and outer ribs **64**, **66** and the pads **68** are designed to engage a perimeter area of the suction cup **28** to control friction during mounting and to hold the suction cup **28** against a mounting surface while suction is being generated by the mount body **12** pulling a center area of the suction cup **28** away from a mounting surface with the threaded post **30**. These features will embed into the material of the suction cup **28** during the mounting operation. The annular rib **58**, inner and outer ribs **64**, **66** and pads **68** may be selected and positioned depending on the friction generated and the holding needs of a particular suction mount. For instance, the annular rib **58** may be discontinuous, the inner and outer ribs **64**, **66** may be not be present on all of the longitudinal ribs **60** or may be only the inner or outer ribs **64**, **66** are employed. The pads **68** could be added to additional longitudinal ribs or reduced in number or even eliminated. Also, the number of longitudinal ribs **60** could be increased or reduced. One benefit of the longitudinal ribs **60** is to reduce material in the mount body **12** and enable the inner and outer ribs **64**, **66** to be intermittent. A pocket **70** is created between adjacent longitudinal ribs **60**.

With reference to FIGS. **5**B and **5**C, there is illustrated an alternate embodiment of a body mount **12'**. The body mount **12'** is almost identical to the body mount **12** (common elements will be denoted with the same reference numbers along with an apostrophe) with the primary difference being an annular wall **59** connecting the longitudinal ribs **60'**. Another difference is that a terminal end of the annular wall **59** includes an annular rib **61** projecting axially away from the wall **59**. The rib **61** replaces the inner ribs **64** of the body mount **12** and provides more friction on the material of suction cup **28** because it is continuous. The wall **59** also closes the inner side of the pockets **70'**. Unlike the body mount **12**, none of the longitudinal ribs **60'** of the body mount **12'** include a pad **68**. As with the body mount **12**, the ribs **58'**, **61**, and **66'** will embed into the material of the suction cup **28** during the mounting operation. The ribs may be selected and positioned depending on the friction generated and the holding needs of a particular suction mount as explained further above for the body mount **12**.

As shown in FIG. **6**, the suction assembly **14** includes the suction cup **28**, the threaded post **30** and the base **32**. As discussed above, the base **32** is positioned in a cavity **71** of the suction cup **28**. The base **32** includes holes **34** through which material **72** of the suction cup **28** flows into during the molding process so that the suction cup **28** cannot rotate relative to the base **32**. The suction cup **28** includes a perimeter region **74** surrounding a centrally positioned concave region **76**.

To install the suction mount **10**, the mount body **12** and the suction cup assembly **14** may be separated by rotating the mount body **12** counter-clockwise relative to the suction cup assembly **14** to unthread the two from one another. Then, the suction cup assembly **14** may be placed against a mounting surface, and the threaded post **30** may be pushed toward the mounting surface to evacuate most, if not all, of the air from the concave region **76**. Next, the mount body **12** is threaded clockwise onto the threaded post **30** and turned until the pads **68** first engage the perimeter region **74** of the suction cup **28**. The pads **68** maintain the suction cup **28** in place initially. The user continues to turn the mount body **12** clockwise to engage the ribs **60**, **64**, **66** with the perimeter region **74** of the suction cup **28** to embed the ribs in the perimeter region **74** to further resist movement of the perimeter region **74** along the mounting surface as the concave region **76** of the suction cup **28** is drawn away from the mounting surface to increase the vacuum force in the concave region **76**. Eventually, the entire annular end surface **56** of the mount body **12** engages the perimeter region **74** of the suction cup **28** with the pads **68** embed deeper into the perimeter region **74** of the suction cup **28** than the ribs **60**, **64**, **66**. The pull force of the threaded post (**30**) and the pressure asserted by the annular end surface (**56**) work against each other so that when combined like they are in the suction mount **10** they greatly increase the suction force of the surface mount **10**. In one embodiment, it takes about 3-6 clockwise turns of the mount body **12** to mount the suction mount **10** to a mounting surface.

Once enough suction force is generated, the suction mount **10** is firmly affixed to the mounting surface. The suction mount **10** can be removed from the mounting surface by reversing this operation.

Turning to FIGS. **7** and **8**, there is illustrated a bracket **78** designed to mount various accessories to the body mount **12**. The bracket **78** has a generally diamond like configuration with four protrusions **80**, **82** separated by arcuate edges **84**. The bracket **78** contains two mounting holes **86** that are spaced symmetrically to align with the accessory mount holes **16** of the mount body **12**. The mounting holes **86** include an annular countersunk conical region **88** so that heads of mounting screws can be flush with a top surface **90** of the bracket **78**.

In between the mounting holes **86**, there is located a hole **92** used to mount accessories. Two of the protrusions **80** define oval slots **94** also for mounting accessories. The oval holes **94** provide flexibility in mounting accessories. The other two protrusions **82** include radial extending elongated slots **96** for mounting accessories. The bracket **78** defines

pockets **98** running along the backside of the elongated slots **96** for receiving a nut to cooperate with a screw of an accessory. The nut may be moved in the pocket **98** radially to the desired position. The nut may have a slight friction fit with the pocket **98**. In one form, the elongated slots can have length that provides a mounting range between the nuts of at least 27.5 millimeter to 78.7 millimeters. The bracket can be used to mount many different accessories, including a boat cleat and a paddle holder just to name a couple.

With reference to FIGS. **9** and **10**, there is illustrated an adjustable angle mount **100** attached to the suction mount **10**. The adjustable angle mount **100** is used to mount accessories in a manner where the angle can be varied between the accessory and the suction mount **10**. The adjustable angle mount **100** includes a base **102** and an arm **104**. The base **102** is shown as circular with an outer perimeter aligning with an outer perimeter of the mount body **12**. The base **102** defines a pair of mounting holes **106** that are positioned to align with the accessory mount holes **16** of the mount body **12**. The mounting holes **106** can have an annular countersunk portion **108** so that a head of a mounting screw can be flush with a top surface **110** of the base **102**.

The arm **104** extends perpendicularly away from the top surface **110** of the base **102**. The arm **104** includes a stepped end configuration with a flat landing **112** and a serrated face **114** perpendicular to the landing **112**. At the center of the serrated face **114**, there is a mounting hole **116**. The serrations of the serrated face **114** extend radially from the mounting hole **116** and cooperate with serrations on an accessory to secure an angle for the accessory relative to the arm **104**. A terminal end **118** of the arm **104** includes a pocket **120** designed to hold a nut **122** to be used with a fastener, such as a thumb screw, for fastening an accessory to the arm **104**. The pocket **120** can be sized to receive the nut **122** with a friction fit so it remains in the pocket **120** after being inserted. A back surface **124** of the base **102** is flat to mate flush with the flat top surface **36** of the mount body **12**.

In FIG. **11**, there is illustrated a double slotted accessory bracket **126** that can be used to attach accessories having a similar style of complementary bracket, such as products from GoPro, Inc. The bracket **126** includes a stepped arm **128** including an arcuate landing **130** and a serrated face **132** perpendicular to the landing **130**. The arm **128** defines an attachment hole **134** and serrations of the serrated face **132** extend radially from the attachment hole **134**. The other end of the arm **128** includes a pair of slots **136** defined by three walls **138**, **140**, **142**. The walls **138**, **140**, **142** each define a hole **144** that all align with one another. Outer wall **142** defines a pocket **146** that can receive a threaded nut (like nut **122**) in a friction fit fashion. The bracket **126** can cooperate with the adjustable angle mount **100** to attach the bracket **126** to the mount body **12** of the suction mount **10**. The serrated faces **114**, **132** cooperate with one another, and the holes **116**, **134** align so that a thumb screw can secure the two brackets **100**, **126** together at the desired angle. The thumb screw can be loosened to allow the serrations of the serrated faces **114**, **132** to cam over one another to change the relative angle. If the terminal end **118** of the bracket **100** is arcuate, then it can easily move in the arcuate landing **130** of the bracket **126**.

With reference to FIGS. **12** and **13**, there is illustrated a single cup holder assembly **148** mounted to the suction mount **10** with the adjustable angle mount **100**. The single cup holder assembly **148** includes a cup holder ring **150** and mounting arm **152**. If the cup is large enough, it can be placed directly in the cup holder ring **150**. Otherwise, a cup accessory **154**, as shown in FIG. **14**, can be placed in the cup holder ring **150**. The cup accessory **154** is shaped like a cup and includes a bottom **156**, an annular side wall **158** and an upper rim **160**. The side wall **158** has a diameter that is smaller than an inner diameter of the cup holder ring **150** and may be tapered. The rim **160** includes an annular flange **162** that has an outer diameter larger than the inner diameter of the cup holder ring **150**. The flange **162** includes a bottom surface **164** that rests on an upper surface **166** of the cup holder ring **150** to support the cup accessory **154**. An inner surface **168** of the cup holder ring **150** may include ribs **170** that engage the side wall of the cup accessory **154** to control the amount of friction engagement between the cup holder ring **150** and the cup accessory **154**.

The mounting arm **152** is stepped and includes an arcuate landing **172** and a serrated face **174** perpendicular to the landing **172**. The arm **152** defines an attachment hole **176** and serrations of the serrated face **174** extend radially from the attachment hole **176**. The other end of the arm **152** is attached to the cup holder ring **150**. The arm **152** could be molded as a single piece with the cup holder ring **150** or could be two pieces attached together using welding, adhesive or other known techniques. When molded as a single piece, a pocket **178** may be added at a transition between the cup holder ring **150** and the mounting arm **152** to reduce the amount of material using to make the single cupholder accessory **148**. The pocket **178** is bordered by the cup holder ring **150**, the mounting arm **152** and three walls **180**, **182**, **184** interconnecting the cup holder ring **150** to the mounting arm **152**.

The mounting arm **152** can cooperate with the adjustable angle mount **100** to attach the mounting arm **152** to the mount body **12** of the suction mount **10**. The serrated faces **114**, **174** cooperate with one another, and the holes **116**, **174** align so that a thumb screw can secure the mounting arm **152** and the bracket **100** together at the desired angle. The thumb screw can be loosened to allow the serrations of the serrated faces **114**, **174** to pass by one another to change the relative angle. If the terminal end **118** of the bracket **100** is arcuate, then it can easily move in the arcuate landing **172** of the mounting arm **172**.

Referring to FIGS. **15-19**, there is illustrated a multi-cup holder accessory **186** mounted to the suction mount **10** with the adjustable angle mount **100**. The multi-cup holder accessory **186** has a body **188** with generally boomerang shape defining three cup holder openings **190**, **192**, **194** and a mounting arm **196**. If the cup is large enough, it can be placed directly in the cup holder openings **190**, **192**, **194**. Otherwise, the cup accessory **154**, as shown in FIG. **14**, can be placed in the cup holder openings **190**, **192**, **194**. The bottom surface **164** of the flange **162** of the cup accessory **154** can rests on an upper surface **198** of the body **188** around each of the cup holder openings **190**, **192**, **194**. Alternatively, the cup **154** can be locked in the center cup holder opening **192** with a pair of locking arms **204**, as discussed further below. An inner surface **200** of the body **188** defining each of the cup holder openings **190**, **192**, **194** may include ribs **202** that engage the side wall of the cup accessory **154** to control the amount of friction engagement between the body **188** at the cup holder openings **190**, **192**, **194** and the cup accessory **154**.

The body **188** may define other openings to hold additional items. These other openings can be sized as desired to hold items used for a specific activity. By way of example only, the additional openings could be designed to hold items commonly used in fishing. In this regard, the body **188** could define an oval opening **206** between two of the cup

holder openings **190**, **192** that would receive a portion of a pliers to support the pliers. The body **188** may also define a pair of elongated slots **208**, **210** of different lengths between two openings **192**, **194** to receive in part different sized knives to hold the knives. Along the front lip **212**, the body **188** may define elongated, arcuate openings **214** for supporting fishhooks. The underside of the body **188** includes an array of recesses to reduce the amount of material used in making the body **188**. The body **188** includes an annular wall **190***a*, **192***a*, **194***a*, **206***a*, **208***a*, **210***a*, **214***a* at each of the openings **190**, **192**, **194**, **206**, **208**, **210**, **214**, respectively, to provide support for and depth to the openings **190**, **192**, **194**, **206**, **208**, **210**, **214** so that the items mounted in the openings **190**, **192**, **194**, **206**, **208**, **210**, **214** do not lean too much.

The mounting arm **196** is stepped and includes a flat landing **216** and a serrated face **218** perpendicular to the landing **216**. The arm **196** defines an attachment hole **220** and serrations of the serrated face **218** extend radially from the attachment hole **220**. The other end of the arm **196** is attached to the body **188**. The arm **196** could be molded as a single piece with the body **188** or could be two pieces attached together using welding, adhesive or other know techniques. When molded as a single piece, pockets **222** may be added at a transition between the body **188** and the mounting arm **196** to reduce the amount of material used to make the multi-cup accessory **186**. The pockets **222** are bordered by the body **188**, the mounting arm **196** and a number of walls **224** interconnecting the body **188** to the mounting arm **196**.

The mounting arm **196** can cooperate with the adjustable angle mount **100** to attach the mounting arm **196** to the mount body **12** of the suction mount **10**. The serrated faces **114**, **218** cooperate with one another, and the holes **116**, **220** align so that a thumb screw can secure the mounting arm **196** and the angle mount **100** together at the desired angle. The thumb screw can be loosened to allow the serrations of the serrated faces **114**, **218** to move pass one another to change the relative angle.

An additional mounting plate **226** may be added to depend from the body **188** if desired. The mounting plate **226** includes a rectangular base **228** that fits into a complimentary rectangular pocket **230** at the underside of the body **188**. The base **228** includes a mounting hole **232** and a pocket **234** that holds a nut **236** in alignment with the mounting hole **232**. The nut **236** may fit in the pocket **234** with a friction fit. The body **188** includes a mounting hole **238** that aligns with the mounting hole **232** of the plate **226** to receive a threaded fastener **240** to attach the plate **226** to the body **188**. The plate **226** includes a front face **242** with two mounting holes **244**. At the rear side, the plate **226** defines two pockets **246**, each to hold a nut **248** in alignment with its respective mounting hole **244**. The nut **248** may fit in its pocket **246** with a friction fit. The spacing between the mounting holes **244** may be designed to mount a specific accessory, including, for example, a conventional bottle opener. The rear of the mounting plate **226** may include recesses **250** to reduce the amount of material.

As mentioned above, the center cupholder opening **192** may include locking arms **204** to attach the cup **154** (FIG. **14**) to the body **188** so that it cannot be unintentionally dislodged. Each locking arm **204** includes a locking portion **252** and a tab **254**. The locking portion **252** includes an arcuate channel **256** bounded by a lower rib **258** and upper wall **260**. The lower rib **258** includes a cam surface **262** so that the cup **154** can be inserted into the opening **192** from underneath the body **188**. As the cup **154** is inserted into the locking arms **204**, the lower portions **252** pivot outward until the flange **162** of the cup **154** passes into the channel **256**. The outward movement of the lower portions **252** is caused by the camming of the flange **162** over the cam surface **262**. The locking arms **204** are connected to the body **188** by relatively small extensions **264** of material that allow the locking arms **204** to pivot and elastically return to a holding position. The body **188** defines an arcuate opening **266** to provide clearance for the locking arm **204** to pivot. The tabs **254** are used to release the cup **154** from the locking arms **204**. The tabs **254** are squeezed towards one another to pivot the channel **256** of the locking portion **252** away from the flange **162** of the cup **154**. This movement allows the cup **154** to drop out of the locking engagement with the locking arms **204**.

As shown in FIGS. **20-23**, there is another multi-cup accessory **268** identical to the multi-cup accessory **186** except that this multi-cup accessory **268** does not include the mounting arm **196**. This multi-cup accessory **268** can be used with the suction mount **10** and the cup accessory **154** to mount the multi-cup accessory **268** on horizontal surfaces, as described further below. The same reference numbers are used for this multi-cup accessory **268** for similar components of the previously described multi-cup accessory **186**.

The cup accessory **154** includes a bottom wall **270** defining a pair of mounting holes **272**. Each mounting hole **272** includes a countersunk portion **274** so that heads of the screws can sit flush with the bottom wall **270**. An underside **276** of the bottom wall **270** may include a boss **278** around each of the mounting holes **272** and a raised perimeter region **280** around its circumference. The mounting holes **272** of the bottom wall **270** align with the mounting holes **16** of the mount body **12** to attach the cup accessory **154** to the suction mount **10**. The cup accessory **154** locks into the center cup holder opening **192** using the locking arms **204**, as described above.

The components described above may be made from a plastic or metal material. For example, the suction cup could be made of all plastic so that it does not rust. In this case, the suction cup may be made of polyvinyl chloride, and the threaded post and the mount body may be made of acrylonitrile butadiene styrene.

While there have been illustrated and described particular embodiments of the present invention, it will be appreciated that numerous changes and modifications will occur to those skilled in the art, and it is intended for the present invention to cover all those changes and modifications which fall within the scope of the appended claims.

What is claimed is:

1. A suction mount system comprising:

a suction cup having a first perimeter region; and

a one-piece body rotatably connected to the suction cup and having a rib extending from a second perimeter region that movably engages and embeds into a material of the suction cup at the first perimeter region of the suction cup when the rib is rotated with the one-piece body to resist movement of the first perimeter region along a mounting surface as a center portion of the suction cup is pulled away from the mounting surface as the suction cup is mounted to the mounting surface,

wherein the one-piece body comprises an outer annular wall having an exterior side and an interior side, and a plurality of longitudinal ribs extend lengthwise along the interior side substantially parallel to a central longitudinal axis of the one-piece body, at least one of the

plurality of longitudinal ribs including at least one ridge at an end thereof to embed into the suction cup.

2. The suction mount system of claim **1** wherein the ridge is a discontinuous annular ridge.

3. The suction mount system of claim **1**, wherein the rib is a first continuous annular rib that embeds into the material of the suction cup at the first perimeter region of the suction cup.

4. The suction mount system of claim **3** further comprising a second continuous annular rib that embeds into the first perimeter region of the suction cup.

5. The suction mount system of claim **1** further comprising a threaded shaft mechanically connecting the suction cup to the one-piece body such that rotating of the one-piece body embeds the rib into the material of the suction cup at the first perimeter region of the suction cup.

6. The suction mount system of claim **5** further comprising a base at least partially molded into the suction cup and attached to the threaded shaft to attach the threaded shaft to the suction cup.

7. The suction mount system of claim **6** wherein the one-piece body defines a hole that is at least partially threaded and receives the threaded shaft.

8. The suction mount system of claim **1** wherein the one-piece body defines mounting holes for attaching accessories to the one-piece body.

9. The suction mount system of claim **8** further comprising a threaded nut at each of the mounting holes.

10. The suction mount system of claim **9** further including a bracket that mounts to the one-piece body using the mounting holes.

11. The suction mount system of claim **10** wherein the bracket defines a plurality of holes.

12. The suction mount system of claim **11** wherein at least two of the plurality of holes are elongated slots.

13. The suction mount system of claim **10** wherein the bracket includes at least two protrusions extending away from one another.

14. The suction mount system of claim **10** wherein the bracket includes a serrated face for mounting accessories at different angles relative to the one-piece mount body.

15. The suction mount system of claim **1** wherein the one-piece body defines a recess and a magnet resides in the recess.

16. The suction mount system of claim **1** further comprising a first accessory defining at least one hole for supporting an object.

17. The suction mount system of claim **16** wherein the at least one hole includes a plurality of holes for supporting different sized objects at the same time.

18. The suction mount system of claim **16** further comprising a second accessory supported by the first accessory and defining a plurality of holes for supporting different sized objects at the same time.

19. The suction mount system of claim **18** further comprising a third accessory supported by the second accessory for supporting at least one object.

20. A method for attachment to a surface comprising:

providing a suction cup having a first perimeter region and a one-piece mount body rotatably connected to the suction cup and the mount body having a rib extending from a second perimeter region;

placing the suction cup against a surface;

rotating the one-piece body; and

movably engaging and embedding the rib into a material of the suction cup at the first perimeter region of the suction cup when the rib is rotated with the one-piece body to resist movement of the first perimeter region along a mounting surface as a central portion of the suction cup is pulled away from the mounting surface as the suction cup is mounted to the mounting surface,

wherein the rib is a discontinuous annular rib and the mount body further includes a first continuous annular rib and a second continuous annular rib, and further comprising the step of embedding the discontinuous annular rib, the first continuous annular rib and the second continuous annular rib into the material of the suction cup at the first perimeter region of the suction cup upon rotating of the mount body.

21. The method of claim **20** further comprising the step of providing a threaded shaft mechanically connecting the suction cup to the mount body such that rotating of the mount body embeds the discontinuous annular rib, the first continuous annular rib and the second continuous annular rib into the material of the suction cup at the first perimeter region of the suction cup.

22. The method of claim **20** further comprising the step of attaching an accessory bracket to the mount body.

23. The method of claim **22** wherein the bracket includes a serrated face for mounting accessories at different angles relative to the mount body.

24. The method of claim **22** further comprising the step of attaching a first accessory defining at least one hole for supporting an object to the bracket.

25. The method of claim **24** wherein the at least one hole includes a plurality of holes for supporting different sized objects at the same time.

26. The method of claim **24** further comprising the step of attaching a second accessory to the first accessory and the second accessory defining a plurality of holes for supporting different sized objects at the same time.

27. A mount for a suction mount system comprising:

a one-piece body,

the one-piece body comprising at least one rib projecting from the one-piece body to engage a suction cup, wherein the at least one rib is fixed against movement relative to the one-piece body such that the rib rotates with the one-piece body when the one-piece body is rotated to engage the suction cup; and

a core of the one-piece body defining a hole that is at least partially threaded to rotatably couple the one-piece body to a threaded portion of the suction cup so rotation of the one-piece body drives axial movement of the threaded portion of the suction cup, and defining an axis about which the one-piece body rotates,

wherein the one-piece body comprises an outer annular wall having an exterior side and an interior side, and a plurality of longitudinal ribs extend lengthwise along the interior side substantially parallel to the axis, at least one of the plurality of longitudinal ribs including at least one ridge at an end thereof to embed into the suction cup.

28. The mount of claim **27** wherein the at least one ridge is discontinuous.

29. The mount of claim **27** wherein the at least one rib is a first continuous annular rib projecting from the one-piece body to engage the suction cup.

30. The mount of claim **29** further comprising a second continuous annular rib projecting from the one-piece body to engage the suction cup.

31. The mount of claim **27** wherein the one-piece body defines mounting holes for attaching accessories to the one-piece body.

32. The mount of claim 28 further comprising at least one pocket formed by the one-piece body adjacent the discontinuous ridge.

33. The mount of claim 27 wherein the one-piece body defines at least one recess to enhance gripping for rotating the one-piece body about the axis.

34. The suction mount system of claim 1 wherein a diameter of the one-piece body at the second perimeter region is less than or equal to a diameter of the suction cup at the first perimeter region.

35. The method of claim 20 wherein a diameter of the one-piece body at the second perimeter region is less than or equal to a diameter of the suction cup at the first perimeter region.

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